



SHZVG Type Oil-Immersed Vacuum On-Load Tap-Changer

Instruction Manual

HM 0.460.7202-08.19/2024

*Important Notice:
Please read this manual carefully
before operating the on-load
tap-changer and keep it for future
reference.*



Shanghai Huaming Power Equipment Co., Ltd.

Preamble

Thank you for choosing our products.

Please read this manual carefully before operating the equipment. Product specifications are subject to change without prior notice. Please refer to the quotation and order confirmation for binding information. Shanghai Huaming Power Equipment Co., Ltd. (hereinafter referred to as "Huaming") shall not be liable for any losses caused by misinterpretation of this manual.

Please keep this manual for future reference. Should you have any questions, please do not hesitate to contact Huaming.

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1. Reading Guide

This chapter provides an overview of the product and explains how to interpret the information presented throughout the manual.

1.1 Document Overview

This technical document provides a detailed description of the operation, monitoring, troubleshooting, and maintenance of the SHZVG On-Load Tap-Changer (OLTC).

It also includes safety instructions and general information relevant to the product.

This document is intended for qualified personnel with specialized training in electrical equipment.

1.1.1 Manufacturer

This product is manufactured by:

Shanghai Huaming Power Equipment Co., Ltd.

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Further information about the product and a copy of this technical document can be obtained by contacting us via the provided email address. Information is subject to change without notice.

The specifications contained in this document are based on the latest information available at the time of publication. Significant revisions will be incorporated into new editions of this manual.

1.1.2 Related Documents

This technical document is a general guide and should be used in conjunction with the following documents:

- Factory Test Report
- Wiring Diagram
- Order Confirmation
- Supplementary Documents

Reading this manual in isolation may not provide a complete understanding of the product and its operation.

1.1.3 Document Retention

Please keep this manual and all related documents in a safe place for future reference.

1.2 Information Indicators

1.2.1 Hazard Warnings

Hazard warnings in this document follow the format below:



Warning label / Hazard type!

Source of hazard / Potential consequences. (Optional)

- Safe practices / Precautions.

Warning Label	Hazard Level	Consequences of Violations
Danger	Imminent hazard	Death or serious injury
Warning	Potential hazard	Severe or fatal injury
Prohibition	Potential hazard	Moderate or severe injury
Caution	Potential hazard	Damage to equipment or personnel

Table 1-1 Warning Labels

Warning Sign	Meaning
	Indicates a potential threat to life and health if the instruction is ignored. Ignoring this warning may result in serious or fatal injury.
	Indicates a risk of electric shock to personnel or damage to equipment if the instruction is ignored. Ignoring this warning may result in serious or fatal injury.
	Indicates a risk of fire, explosion, or other thermal hazards if the instruction is ignored.
	General prohibition sign. The red symbol indicates that the action is forbidden.

Table 1-2 Symbols Used for Warning Signs

1.2.2 Important Notes

The following symbol is used throughout this document to highlight important information:



Summary/Type/Description !

This blue annotation symbol signifies important information that requires attention.

1.2.3 Procedural Instructions

Procedural instructions are presented as single-step or multi-step operations.

1.2.3.1 Single-Step Operations

A single-step operation consists of a single action. The format is as follows:

Objective:

- Pre-operation preparation. (Optional)

Step:

- Specific instructions / Expected results. (Optional)
- Outcome. (Optional)

1.2.3.2 Multi-Step Operations

A multi-step operation consists of multiple sequential actions. The format is as follows:

Objective:

- Pre-operation preparation. (Optional)

1. Step 1

- Specific instructions / Expected results. (Optional)

2. Step 2

- Specific instructions / Expected results. (Optional)
- Outcome. (Optional)

2. Pre-Operational Preparations

Thoroughly read this manual to familiarize yourself with the SHZVG OLTC.

This manual is an integral part of the product.

Carefully review and adhere to the safety instructions presented in this chapter.

Pay close attention to all warning information throughout this manual to prevent operational hazards.

This product is designed and manufactured using state-of-the-art technology. However, improper operation can pose a risk to personnel, damage the equipment, or result in property loss.

2.1 Basic Safety Precautions

2.1.1 Safety Measures

Precaution	Detailed Instructions
Electric Shock	Always disconnect the equipment from the power supply before inspection or maintenance. Maintain a safe distance from high-voltage equipment and wear appropriate protective clothing.
Transformer Oil	Collect used transformer oil in designated containers. Keep oil away from open flames or sparks. Avoid direct contact with transformer oil, as it can be harmful to health.
Fire	In case of fire, use a dry chemical, foam, or carbon dioxide extinguisher. Do not use water.
Waste Disposal	Dispose of waste materials properly according to local regulations. Different waste categories require specific disposal methods. Improper handling of waste can pose a threat to the environment and human health.

Table 2-1 Safety Measures

2.1.2 Intended Use and Compliance

This product, including associated equipment and specialized tools, complies with all applicable laws, regulations,

and standards in effect at the time of delivery, specifically those pertaining to health and safety.

When operated in accordance with the instructions, conditions, and warnings provided in this manual and on the product itself, the SHZVG OLTC does not pose a hazard to personnel, property, or the environment. This applies throughout the product's entire lifecycle, from delivery, installation, and operation to decommissioning and disposal.

Our quality assurance system guarantees a consistently high standard of quality, particularly with regard to health and safety requirements.

The following conditions constitute intended use:

1. The product is operated in accordance with this manual, the agreed upon delivery conditions, technical specifications, and any associated equipment or specialized tools provided are used solely for their intended purpose and within the specifications outlined in this document.
2. This OLTC must only be used with the transformer specified in the order documentation. The serial number of the Motor-Drive Unit (MDU) must correspond to the serial number of the OLTC.

2.1.3 Improper Use

Any use of the product outside the scope defined in Section 2.1.2 is considered improper.

Huaming shall not be held liable for any damages or losses arising from unauthorized modifications or improper use of the product. Unauthorized modifications without prior consultation with Huaming may result in personal injury, damage to the equipment, and operational malfunctions.

2.2 Operator Responsibilities

2.2.1 Operator Qualifications

This product is designed for use in electrical power systems and installations operated exclusively by qualified personnel. Qualified personnel are those who have received appropriate training and are familiar with the installation, assembly, commissioning, and operation of such equipment.

2.2.2 Operator Obligations

To prevent accidents, malfunctions, damage, and adverse environmental impact, personnel responsible for the transportation, installation, operation, maintenance, and disposal of this product or its components must ensure the following:

1. All warnings and hazard notices are strictly observed.
2. Personnel receive regular training on all aspects of operational safety, including operating instructions, with particular emphasis on safety precautions.
3. Safety regulations, operating instructions, and emergency procedures are readily accessible at all times and prominently displayed in the workplace as applicable.

4. The product is operated only in a safe and functional condition. Safety devices, in particular, must be regularly inspected for proper operation.
5. Only manufacturer-authorized replacement parts, lubricants, and auxiliary tools are used.
6. The specified operating conditions and installation requirements are strictly adhered to.
7. All necessary tools and personal protective equipment (PPE) are provided for each task.
8. The prescribed maintenance intervals and procedures are strictly followed.
9. The installation, electrical connections, and commissioning of the product are carried out exclusively by qualified and trained personnel in accordance with this manual.
10. Operators must ensure the proper and intended use of the product at all times.

2.2.3 Personal Protective Equipment (PPE)

Safety is paramount when operating this product. To mitigate potential hazards, the following PPE must be worn at all times during operation and maintenance:

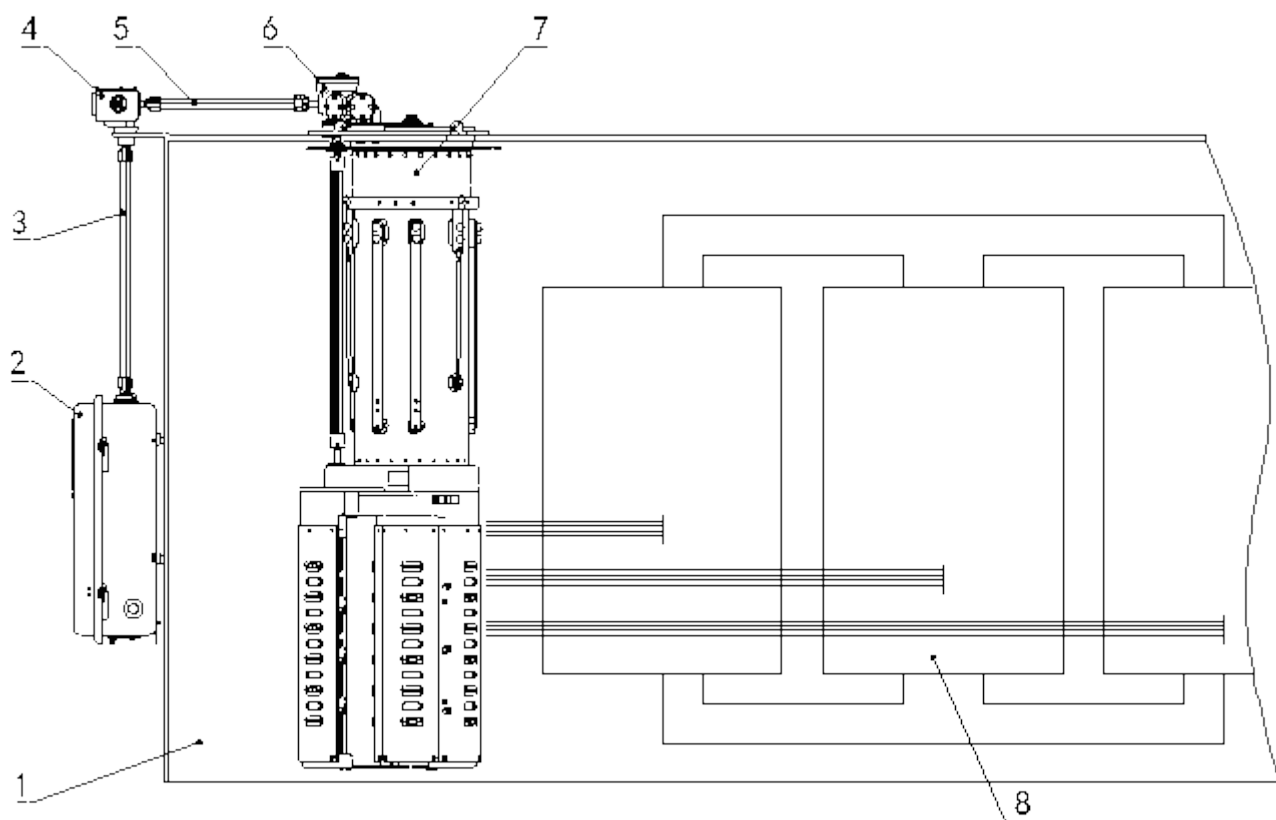
	Wear clothing that provides adequate protection against the hazards present in the work environment.
	Wear safety shoes that offer protection against punctures from sharp objects and provide slip resistance on potentially slippery surfaces.
	Wear safety glasses to protect eyes and face from impact, dust, splashes of liquids, and heat.
	Wear a hard hat to protect the head from impacts and falling objects.
	Wear hearing protection when working in noisy environments to prevent hearing damage.
	Wear gloves that offer protection against mechanical, chemical, and electrical hazards.

Table 2-2 List of Personal Protective Equipment

3. Product Data

3.1 Product Structure

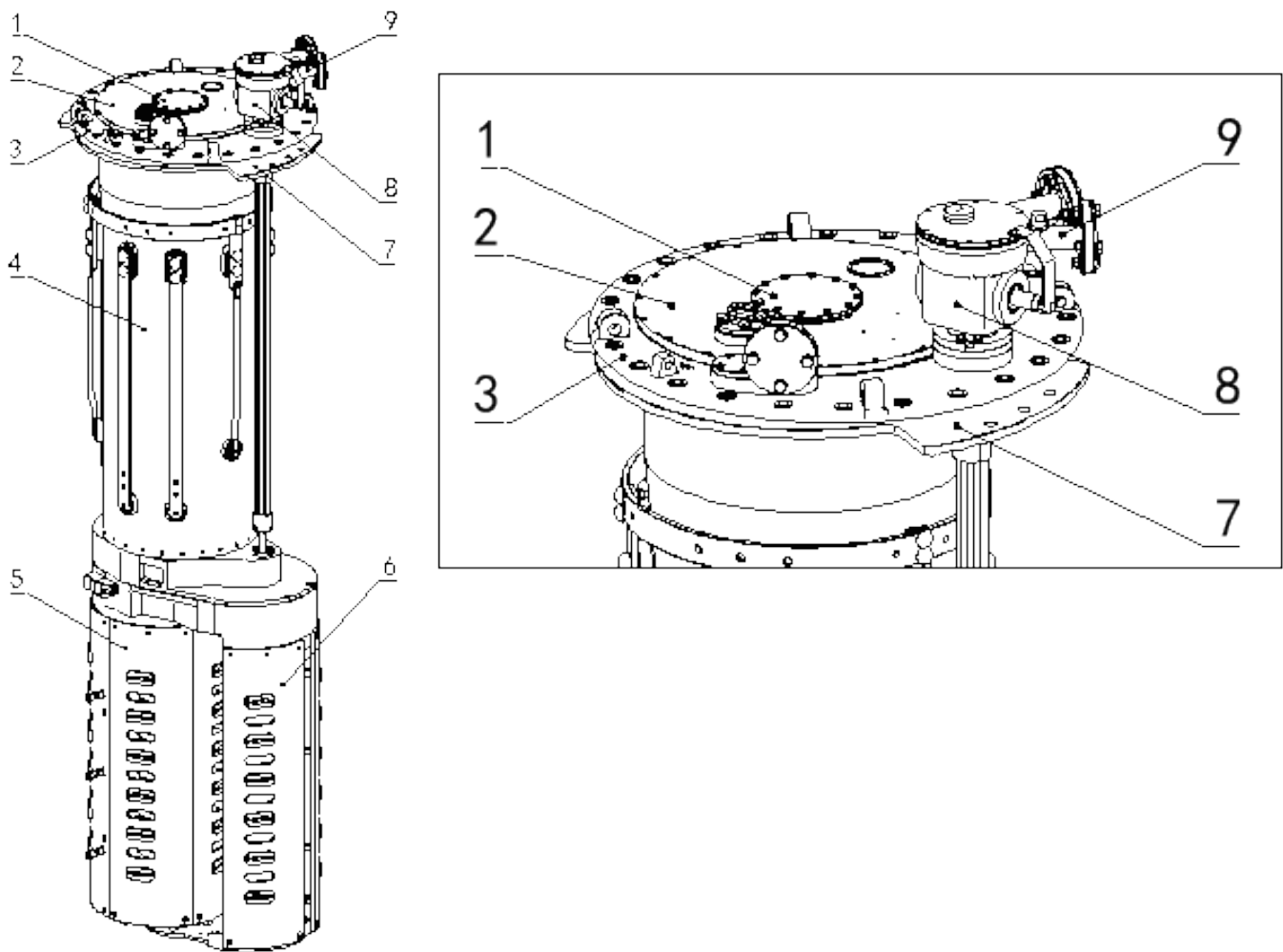
Le changeur de prises en charge à vide à bain d'huile de type SHZVG (ci-après dénommé « OLTC ») est monté sur le couvercle du transformateur à l'aide d'une bride de montage. Il est relié à la MDU par l'intermédiaire d'un réducteur à denture droite, d'un réducteur conique (accessoire) et d'arbres d'entraînement horizontaux et verticaux. Cette configuration permet des opérations de changement de prises motorisées et à distance (voir Figure 3-1).



1. Transformer Tank	2. Motor-Drive Unit (MDU)
3. Vertical Drive Shaft	4. Bevel Gear Box
5. Horizontal Drive Shaft	6. Spur Gear Box
7. OLTC Body	8. Transformer Winding

Figure 3-1 Overview of the OLTC in a Transformer

The SHZVG type oil-immersed OLTC is a combination-type OLTC, consisting of a diverter switch and a tap selector. (See Figure 3-2).



1. Pressure Relief Device	2. Diverter Switch Top Cover
3. OLTC Mounting Flange	4. Diverter Switch
5. Tap Selector	6. Changeover Selector
7. OLTC Support Flange	8. Spur Gear Box
9. Oil Pipe	

Figure 3-2 Overview of the OLTC Structure



Note!

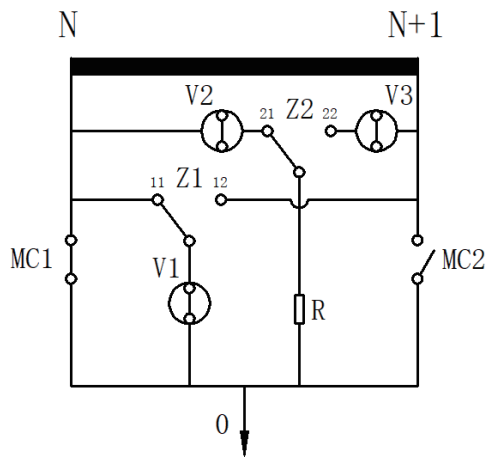
It is recommended to use a protective relay equipped with a gas detection feature.

3.2 Technical Specifications

Table 3-1 Technical Data

Item	Model		SHZVG III 1300Y SHZVG I 1300	SHZVG III 1500Y SHZVG I 1500	SHZVG I 2000	SHZVG I 2400	SHZVG I 3000	SHZVG I 3600
1	Maximum Rated Through-Current (A)		1300	1500	2000	2400	3000	3600
2	Rated Frequency (Hz)		50 or 60					
3	Connection Type		Three-phase Wye / Single-phase (Any)					
4	Maximum Rated Step Voltage (V)		6000					
5	Rated Step Capacity (kVA)		6000		9600		14400	
6	Short-Circuit Withstand Capability (kA)	Thermal (3s)	15	20	24	30	32	40
		Dynamic (Peak)	37.5	50	60	75	80	100
7	Maximum Number of Tap Positions		Without Changeover Selector: 18; With Changeover Selector: 35					
8	Insulation Level to Ground (kV)	Maximum Equipment Voltage	72.5	126	170	252	300	363
		Rated Power Frequency Withstand Voltage (50Hz, 1 min)	140	230	325	460	480	510
		Rated Lightning Impulse Withstand Voltage (1.2/50μs)	350	550	750	1050	1100	1175
9	Tap Selector		Classified as Type B, C, D, DE or E based on insulation level					
10	Mechanical Life		1.5 million operations					
11	Maintenance-Free Operations		300,000 operations					
12	Diverter Switch Oil Compartment	Operating Pressure	0.03MPa					
		Sealing Performance	0.1 MPa, 24 hours leak-free					
		Overpressure Protection	Pressure relief device activates at 0.3 ± 20% MPa					
		Protective Relay	Set flow rate: 2.5 m/s ± 10%					
13	Compatible MDU		SHM series or CMA7					

3.3 Electrical Topology Diagram



MC1, MC2: Main Contacts (Odd and Even Taps)

0: Current Output Terminal

V1: Main Vacuum Interrupter

V2, V3: Transition Vacuum Interrupters

Z1, Z2: Isolating Contacts

R: Transition Resistor

3.4 Operating Conditions

1. The permissible operating temperature range for the OLTC in an oil environment is -25°C to $+105^{\circ}\text{C}$.
2. The ambient temperature at the installation site should be within the range of -25°C to $+40^{\circ}\text{C}$, with a relative humidity not exceeding 85%. For operation outside these parameters, please specify your requirements when placing your order. Special materials and accessories can be incorporated into the OLTC design to accommodate specific environmental conditions.
3. After installation on the transformer, the verticality of the OLTC with respect to the ground should not deviate by more than 2%.
4. The OLTC should be installed in a location free from excessive dust, explosive gases, or corrosive gases.



Environmental Warning !

Condensation can lead to a reduction in electrical performance, potentially damaging the equipment and creating an explosion hazard.

- Do not install the equipment in locations where the relative humidity exceeds 85%..



Caution!

- Customized designs are available for operation in specific environments. Please contact Huaming for further information.

3.5 Model Designation

3.5.1 Model Number Structure

The SHZVG OLTC is available in various configurations based on different combinations of the number of phases, maximum rated through-current, maximum equipment voltage, tap selector insulation class, and connection

type. Therefore, these performance parameters are reflected in the OLTC model designation. Figure 3-3 provides a detailed explanation of each code within the model number.

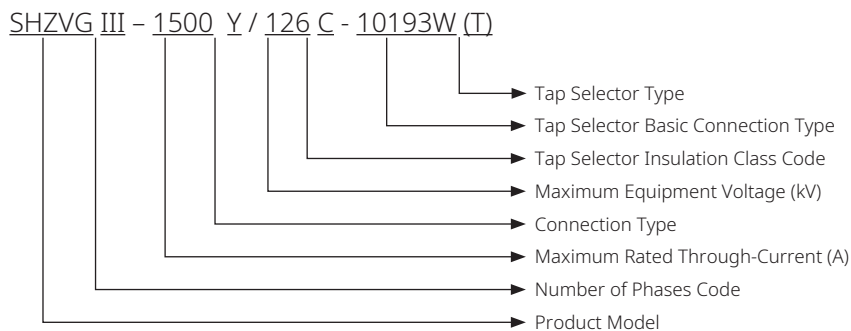


Figure 3-3 Model Number Structure

3.5.2 Basic Connection Types

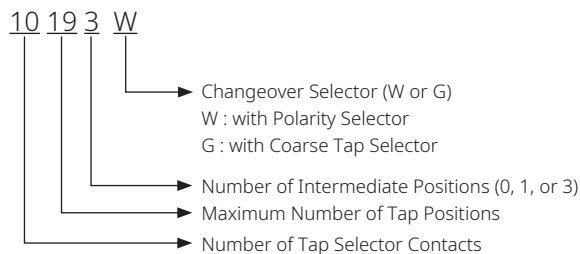


Figure 3-4 Basic Connection Type Designation

1. For example, 10090 indicates a linear tap selector with 10 tap selector contacts and 9 maximum working positions. The "0" signifies that it is a linear tap selector without a changeover selector.
2. For example, 10193W represents a tap selector with 10 tap selector contacts, 19 maximum working positions, and 3 intermediate positions. The "W" indicates the presence of a polarity changeover selector.
3. For example, 10191G represents a tap selector with 10 tap selector contacts, 19 maximum working positions, and 1 intermediate position. The "G" indicates the presence of a coarse tap changeover selector.

3.5.3 Tap Selector Type

The tap selector types are represented by "T" and "L".

"T" designates a barrel-type tap selector and is used when all three of the following conditions are met:

- Number of tap selector contacts does not exceed 14.
- Three-phase rated current does not exceed 1300A.
- Insulation level is not higher than Class D.

"L" designates a cage-type tap selector and is used when one or more of the following conditions are met :

- Number of tap selector contacts exceeds 14.
- Three-phase rated current exceeds 1300A.
- Insulation level is higher than Class D.

4. Packaging, Transportation, and Storage

4.1 Supplied Components

The SHZVG OLTC is supplied as a complete set consisting of the following components:

1. Diverter Switch
2. Tap Selector
3. Motor-Drive Unit (MDU), Controller, and Connecting Cables
4. Drive Shaft, Coupling, Bevel Gear Box, and other Transmission Accessories
5. Protective Relay and Accessories
6. Specialized Tools and Accessories for OLTC Assembly
7. Accessories for Special Requirements Specified by the Customer

Note: The specific components included in the delivery are subject to the packing list provided with the shipment.

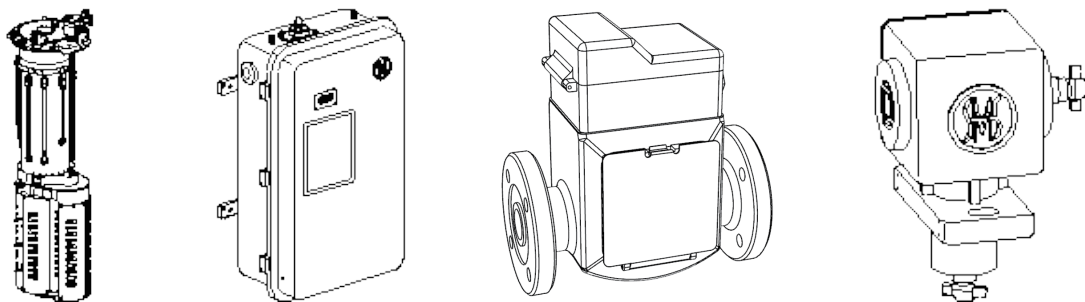
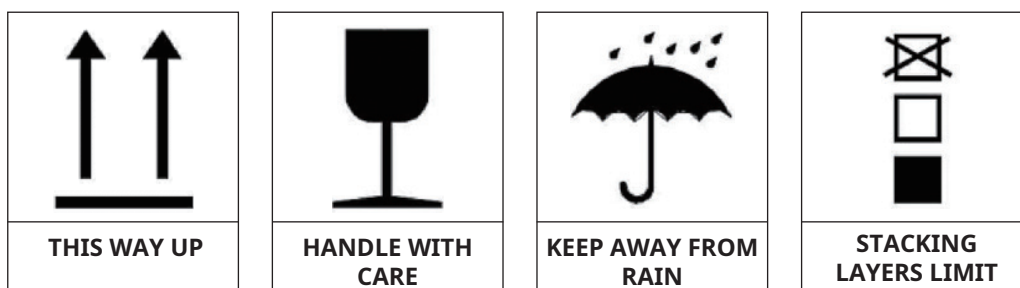


Figure 4-1 Schematic Diagram of Supplied Components (Refer to Actual Product for Details)

4.2 Transportation and Acceptance

4.2.1 Transportation

1. The factory packaging is designed to withstand various modes of transportation. Boxes may be stacked during storage, provided the load on the top surface does not exceed 500 kg/m².
2. During transportation, the boxes should be positioned according to the marked center of gravity. Lifting operations must be carried out in accordance with the lifting points indicated on the packaging.
3. Packaging Warning Labels (See Figure 4-2)



Gross Weight (G.W.) Net Weight (N.W.) KG

Figure 4-2 Packaging Warning Labels

Risk of Toppling!

Toppling or falling packages can cause serious injury to personnel and damage to the equipment.

- Sling selection and load securing must be performed by trained and authorized personnel.
- Do not stand under suspended loads during lifting operations.
- Use transportation equipment and lifting devices with a load capacity greater than 500 kg.

4.2.2 Handling

Upon arrival at the destination, handle the OLTC package according to the unloading instructions, using appropriate methods for transfer.

If the transformer is shipped in a wooden crate, lifting should be performed using the designated lifting points marked on the crate. Alternatively, the crate can be moved using a forklift, as shown in the figures below.

**Caution!**

- Select a forklift and/or crane with appropriate capacity based on the dimensions and gross weight of the crate.

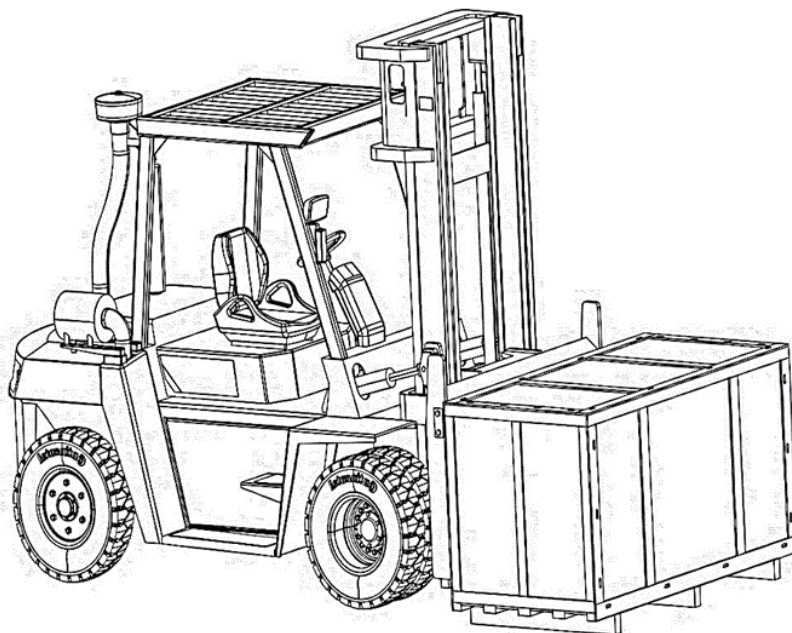


Figure 4-3 Forklift Handling of Wooden Crate

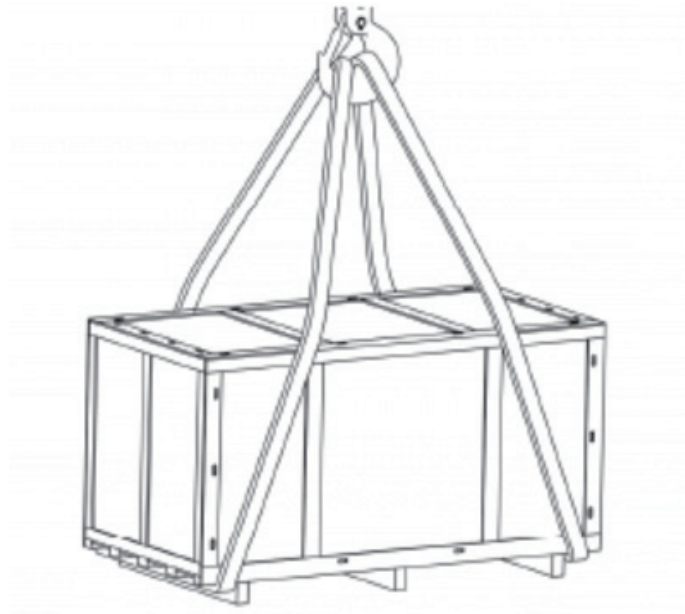


Figure 4-4 Crane Lifting of Wooden Crate

4.2.3 Lifting

Lifting Hazard!



Improper lifting can cause damage to the equipment and injury to personnel.

- During lifting operations, secure at least two horizontal lifting points or all designated lifting points before hoisting.
- Avoid rapid upward or downward movements during lifting.

Personnel are strictly prohibited from standing underneath the load during lifting operations.

The OLTC body should be lifted vertically. Ensure that all lifting points on the OLTC mounting flange are securely attached before lifting. (See Figure 4-5).

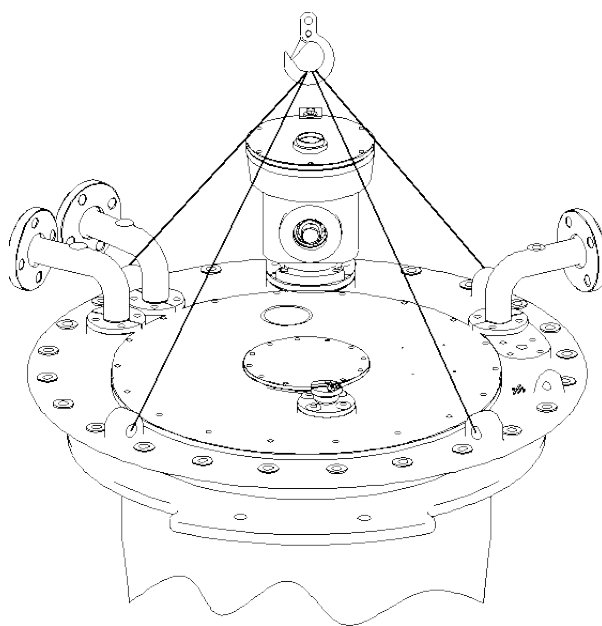


Figure 4-5 OLTC Lifting Diagram

4.2.4 Receiving Inspection

Important Notes!

Unforeseen incidents can occur during transportation. Please pay close attention to the following during the receiving inspection:

- Upon receipt of the shipment, carefully inspect the packaging and all components against the packing list. Immediately report any damage or missing items to Huaming.
- Transportation and lifting of the package must be carried out by qualified personnel.
- The transportation vehicle and lifting equipment must have a minimum load capacity of 1000 kg.
- During transportation and handling, the package must be secured to prevent damage to the OLTC due to vibration, impact, dropping, tilting, collision, or swaying.
- If the package has been dropped or subjected to significant impact, potential damage must be assumed. A thorough inspection of the OLTC is required. If necessary, arrange for an inspection and/or repair by qualified Huaming personnel.



Customer Receiving Inspection Procedures:

Each shipment must be inspected and verified by the recipient before acceptance (Confirmation of Acceptance form). The recipient should verify the following:

1. Verify the quantity of received items against the shipping manifest.
2. Verify that the product specifications match the order documentation.
3. Inspect the packaging for any signs of damage..

If any damage to the packaging is discovered upon receipt (photograph the damaged packaging and contents if possible), or if the contents do not match the shipping manifest, refuse the delivery and immediately contact the carrier to report the discrepancy. If the issue cannot be resolved with the carrier, notify Huaming's After-Sales Service department in writing.

These procedures also apply in cases where the packaged OLTC has been exposed to moisture (rain, snow, or condensation). Store all components in a dry location before installation.

The OLTC must remain in its sealed packaging until immediately prior to installation.

4.2.5 Unpacking and Inspection

Note !



- Take care not to damage the original packaging during unpacking and inspection.
- If the received goods or accessories do not match the packing list, contact Huaming immediately.
- Transport the package to the installation location.
- Verify the equipment and accessories against the packing list..

If the package has been dropped or impacted significantly after delivery, potential damage must be considered. Thoroughly inspect the OLTC and, if necessary, contact Huaming to arrange for an inspection and/or repair by qualified personnel.

4.3 Storage

**Note !**

Pay close attention to the following during storage to prevent damage or deterioration:

- After prolonged storage, the OLTC and its accessories must be thoroughly inspected before installation. Verify that the OLTC specifications match the transformer requirements. Each OLTC is designed for a specific transformer and should not be interchanged.

If the packaged OLTC is stored for more than one year, a detailed inspection is required before installation.

Specially packaged units with sealed packaging can be stored outdoors, provided the following conditions are met:

1. The ambient temperature at the storage location must be within the range of -25°C to +40°C. For applications requiring operation in ambient temperatures from -60°C to +40°C, the OLTC can be manufactured with special materials and accessories to meet low-temperature requirements. Specify this requirement when placing your order.
2. Do not store the OLTC in environments containing flammable, explosive, or corrosive gases.
3. Protect the stored OLTC from moisture, dust, rodents, and insects.
4. Regularly inspect the stored OLTC for any signs of damage or deterioration.
5. If the storage period is extended, replace the desiccant periodically and ensure the packaging remains sealed..

5. Installation of the On-Load Tap-Changer on the Transformer



Installation Notice!

Improper installation can damage the OLTC.

- For equipment rated above 35kV, ensure that the shields provided for the bolts are installed during the installation process.

5.1 Transformer Mounting Flange for Securing the OLTC

The OLTC is mounted to the transformer cover using a dedicated transformer mounting flange. The flange must be designed to accommodate the shape of the sealing surface on the OLTC mounting flange (See Figure 5-1). The 24 M12 bolts must be accurately positioned.

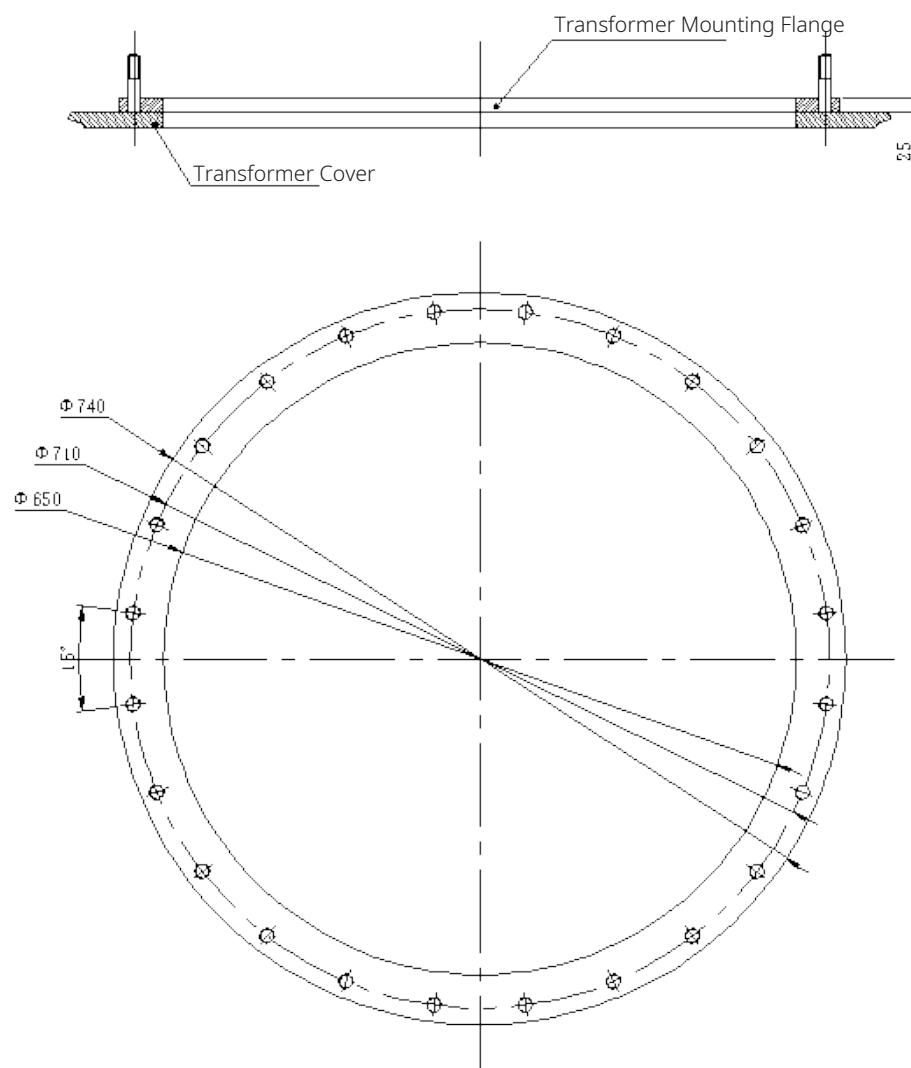


Figure 5-1 Mounting Flange for Securing the OLTC

5.2 Installation of the OLTC on a Bell-Type Transformer

5.2.1 Removing the Top Cover and Lifting Out the Diverter Switch Insert



Risk of Falling!

Improper handling can result in equipment damage and personnel injury.

- Place components on a flat, clean surface.
- Falling components can cause injury.

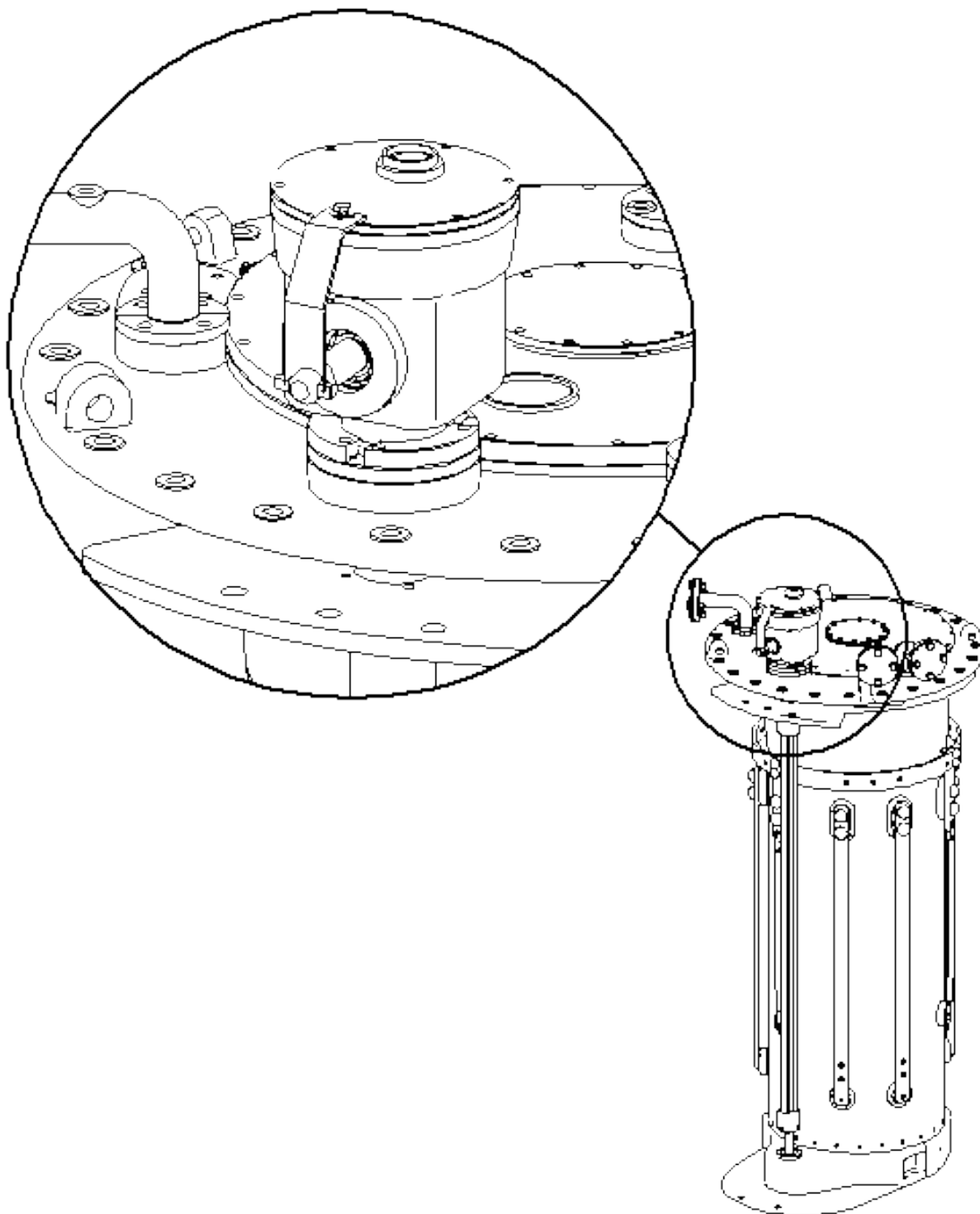


Figure 5-2 Stop Plate Locking Position on the Spur Gear Box

1. Verify that the stop plate on the OLTC spur gear box is secured (see Figure 5-2).
2. Remove the 18 M8 bolts from the top cover (with spring washers, use a 13mm wrench).

**Tip!**

- Keep the top cover gasket in good condition.
- Avoid damaging the sealing surfaces of the top cover and the OLTC mounting flange (O-ring, see Figure 5-3) during disassembly.

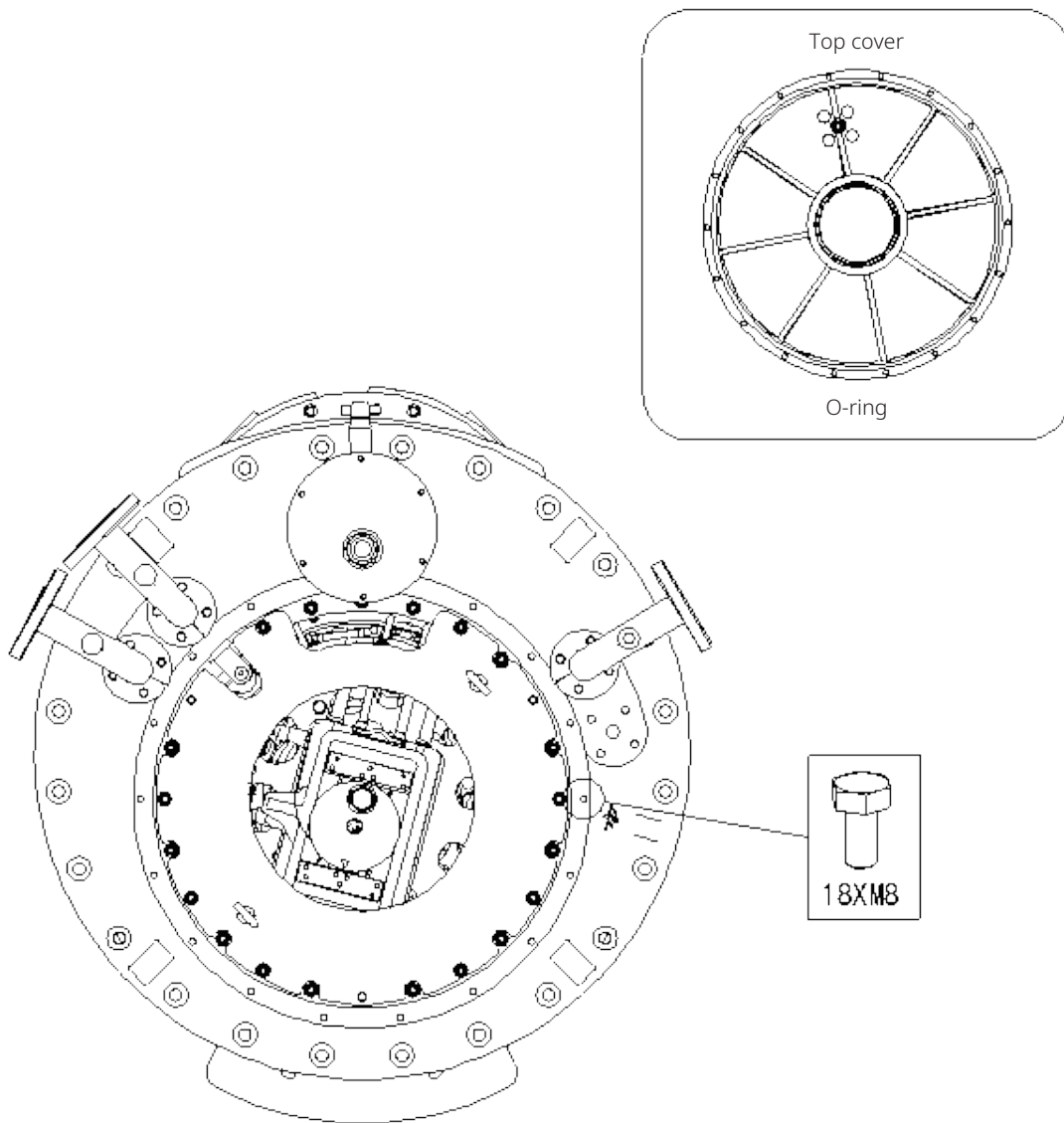


Figure 5-3 Top Cover Removal Diagram

3. Remove the top cover.

Caution !



- Do not damage the O-ring on the pipe fitting.
- Prevent fasteners from falling into the Diverter Switch oil compartment. Keep all fasteners safely stored.
- Tighten the eight M8 nuts when the OLTC is tested with the transformer.

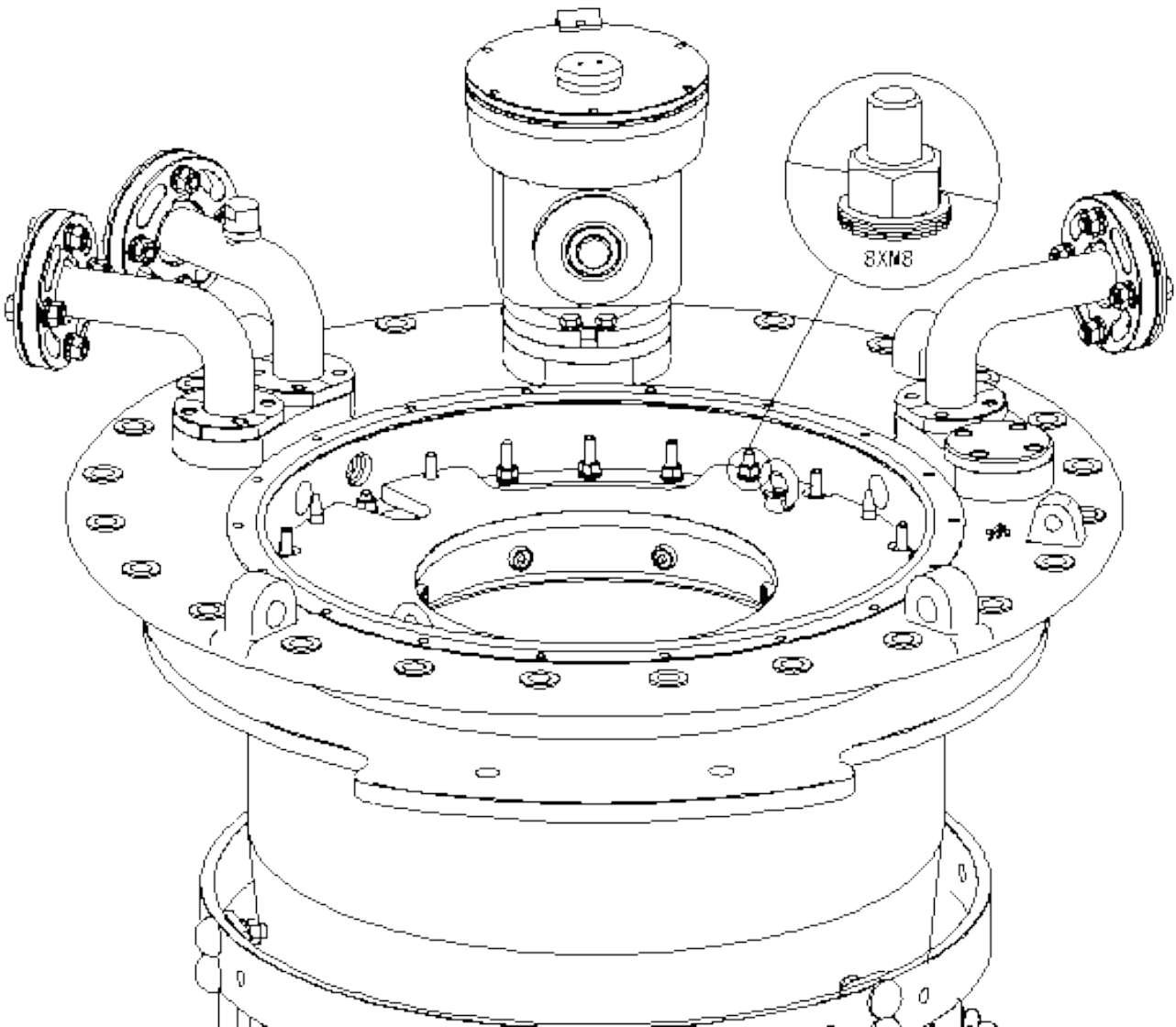


Figure 5-4 Removing the Nuts Securing the Diverter Switch Insert

4. Remove the nuts from the areas on the flange that are not marked with red paint (eight M8 nuts with washers, use a 13mm wrench) (see Figure 5-4).
5. Slowly lift the Diverter Switch Insert vertically out of the oil compartment (see Figure 5-5).

Installation Notice !



Follow these instructions to prevent equipment damage.

- Support both sides of the Diverter Switch Insert during lifting to prevent the drive shaft at the bottom from contacting the ground.
- Do not rotate the Diverter Switch Insert before the Diverter Switch is properly installed.



Caution !

- Place the Diverter Switch Insert on a level surface.
- Prevent the Diverter Switch Insert from tilting.
- Do not change the position of the Tap Selector after removing the Diverter Switch Insert.

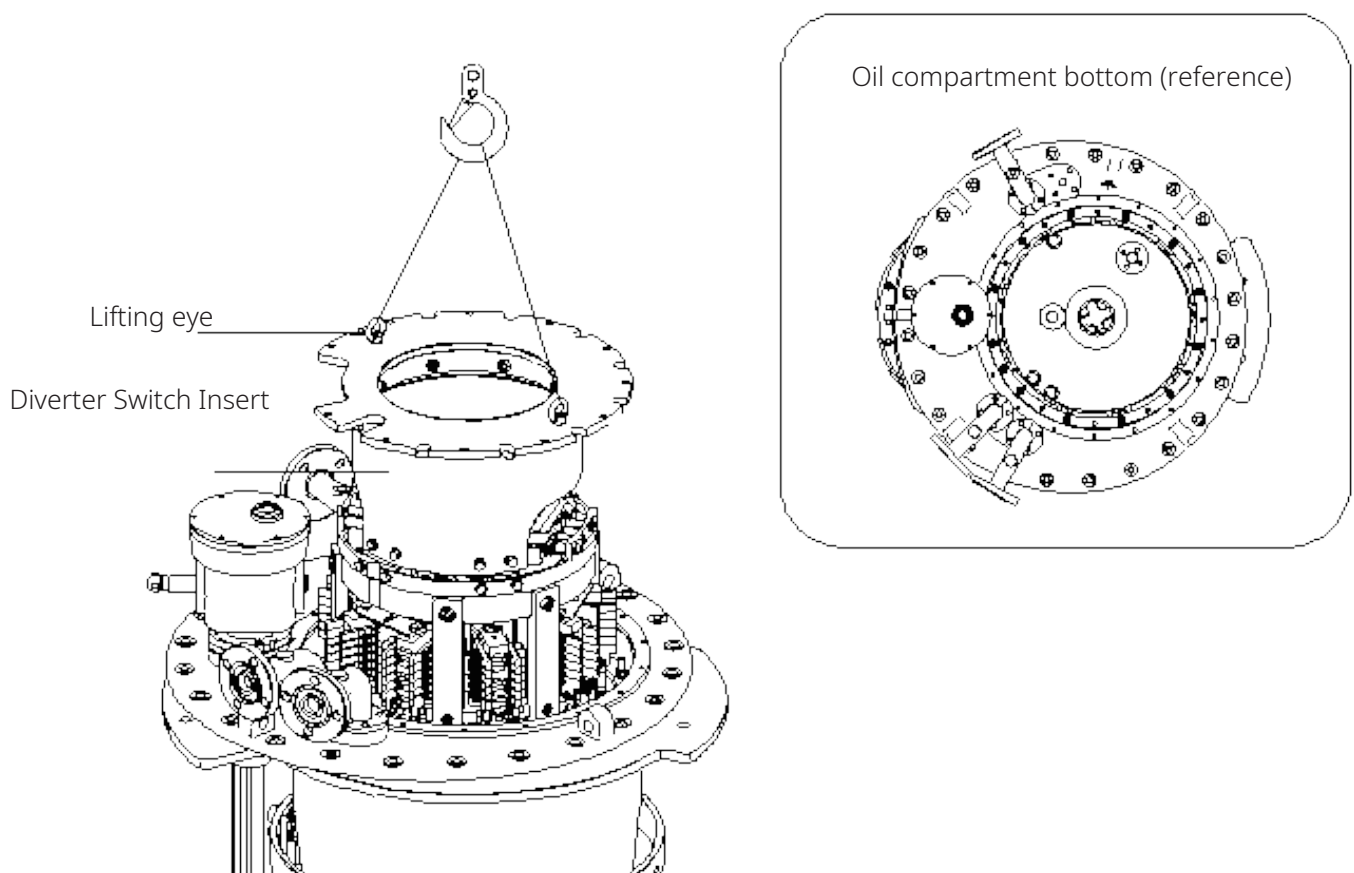


Figure 5-5 Vertically Lifting the Diverter Switch Insert

6. Remove the oil drain pipe (see Figure 5-6).

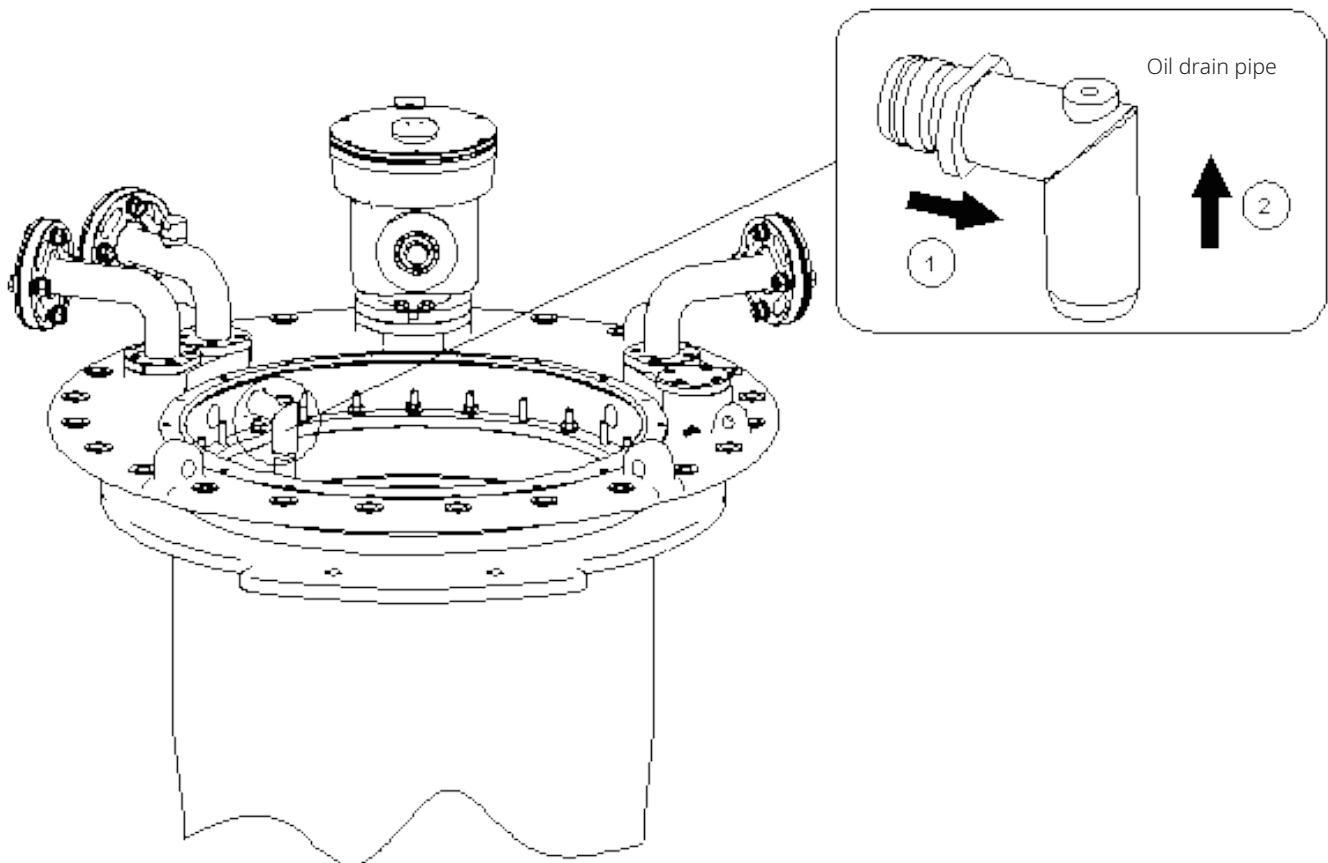


Figure 5-6 Oil Drain Pipe Removal Diagram

5.2.2 Removing the OLTC Mounting Flange



Caution !

Prevent fasteners from falling into the Diverter Switch oil compartment. Keep all fasteners safely stored for reassembly.

1. Remove the remaining nuts from the OLTC mounting flange (thirteen M8 nuts with wing washers, use a 13mm wrench).
2. Lift the mounting flange off the support flange (see Figure 5-7)..

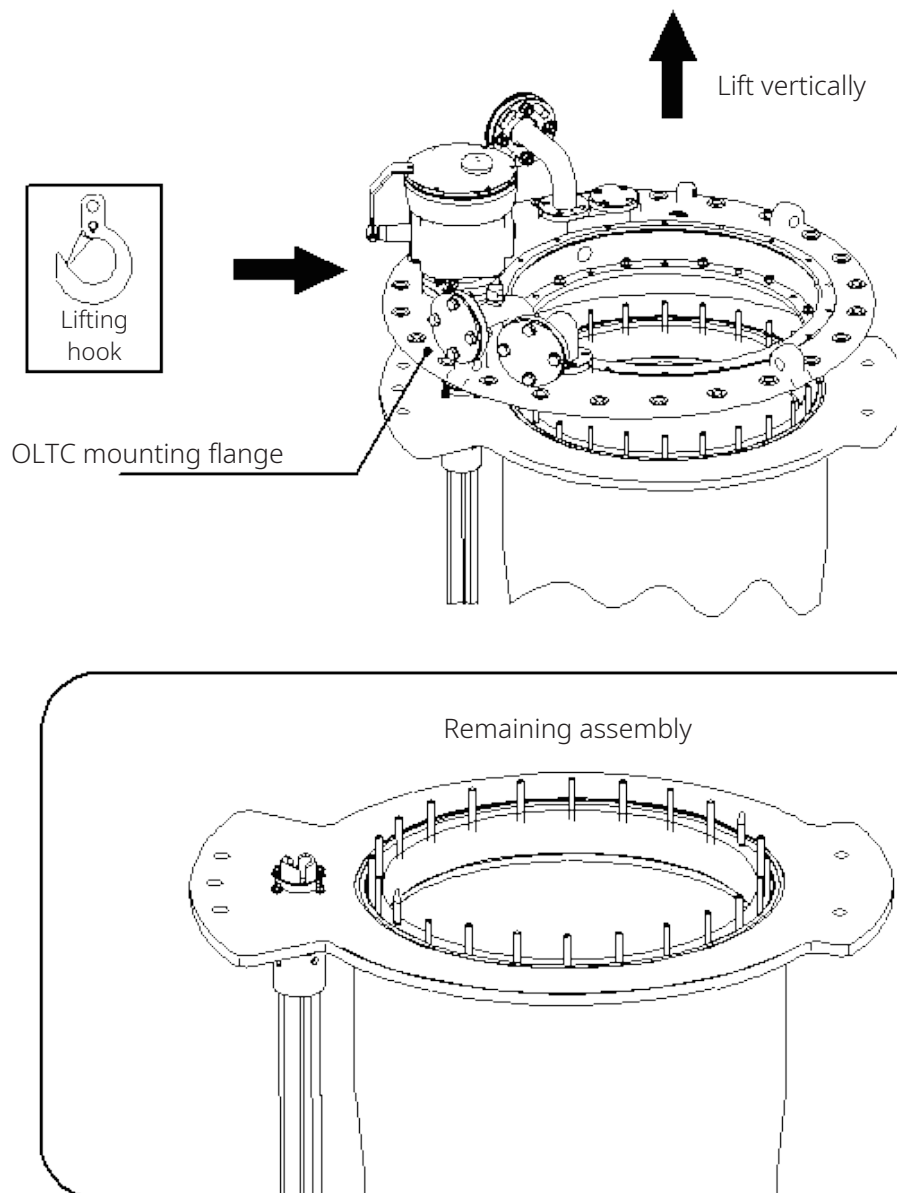


Figure 5-7 Lifting the OLTC Mounting Flange

5.2.3 Assembling the Tap Selector and Diverter Switch Oil Compartment



Operational Caution !

Incorrect connection can severely damage the equipment.

- Do not rotate the splined shaft after removing the locating plate.

Lift the Diverter Switch oil compartment onto the top of the Tap Selector and connect them. Follow these steps:

1. Place the Tap Selector on a level surface.
2. Remove the Tap Selector locating plate (see Figure 5-8).

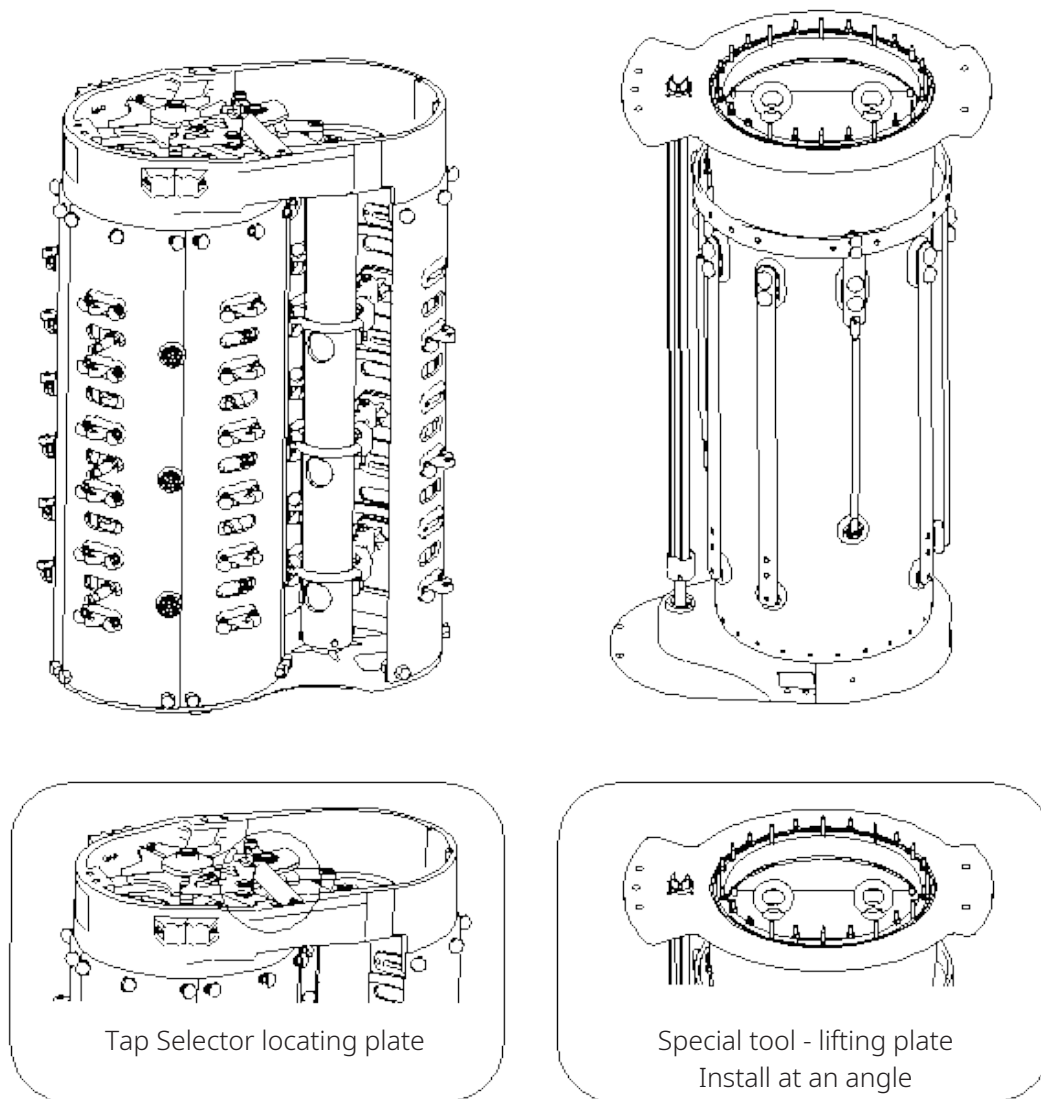


Figure 5-8 Tap Selector Locating Plate and Lifting Ring for the Oil Compartment

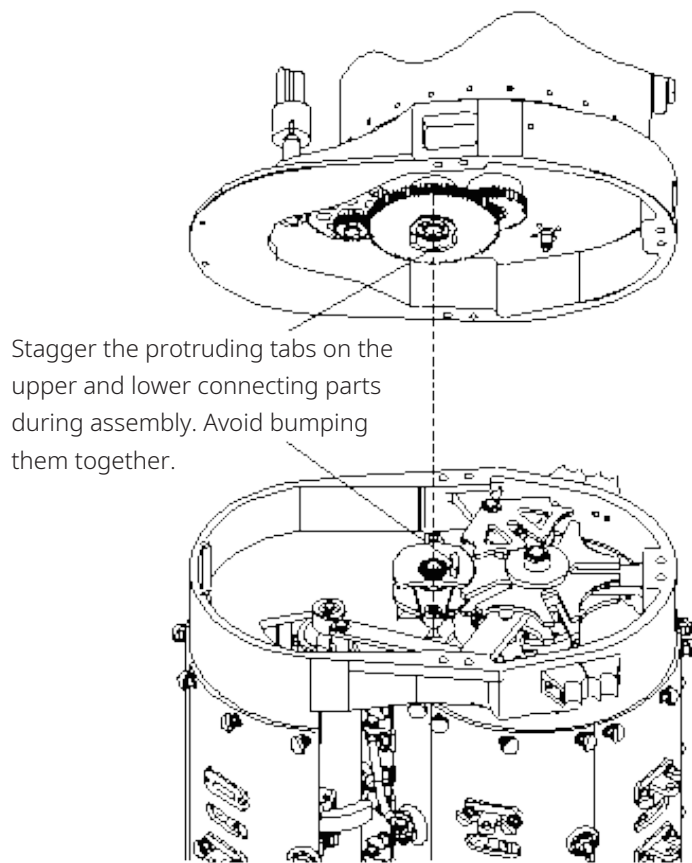
3. Use the special tool to lift the Diverter Switch oil compartment (see Figure 5-8).
4. Carefully lower the oil compartment onto the Tap Selector (see Figure 5-9).

Caution !

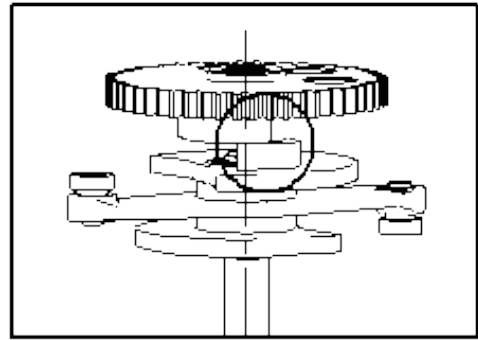
Failure to follow these instructions can damage the equipment.

- Do not rotate the coupling after removing the locating pin.
- Maintain verticality between the Diverter Switch and the Tap Selector (deviation less than 2%).





Incorrect:



Correct:

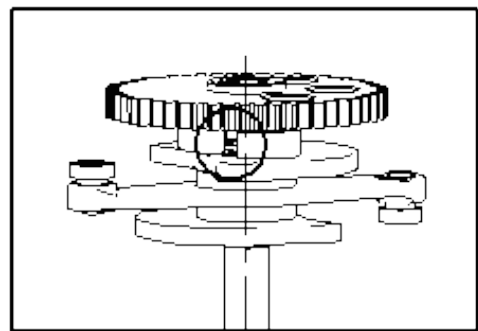


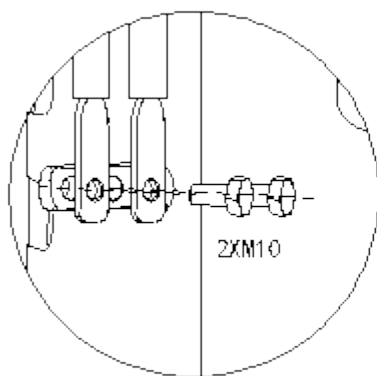
Figure 5-9 Connecting the Tap Selector and Diverter Switch Oil Compartment

5. Align the two couplings. Ensure the protruding tabs on the upper and lower coupling halves are staggered to prevent interference. (See Figure 5-9.)
6. Align the mounting holes on the Diverter Switch oil compartment and the Tap Selector. Use the six M12 bolts and nuts (with washers, 19mm wrench) provided in the Tap Selector packaging accessory box to connect them (See Figure 5-9).
7. Install the odd and even lead wires between the Tap Selector and the Diverter Switch oil compartment:
 - Retrieve the lead wires from the Diverter Switch packaging box.
 - Connect the Diverter Switch terminals to the Tap Selector terminals.
 - Use the M10 x 35 bolts, nuts, washers, shields, and a 17mm wrench to connect the terminals.
8. Après avoir connecté les fils conducteurs avec des boulons, fixez les protections.

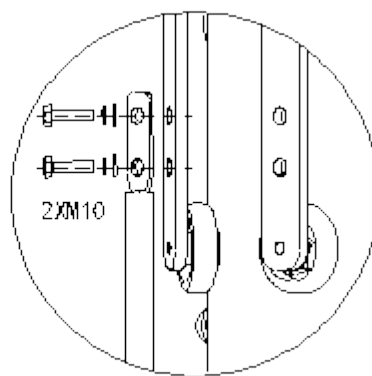
Caution !

Follow these installation points to prevent equipment damage.

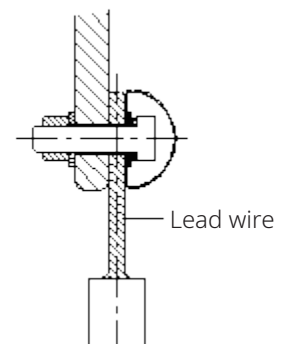
- Connect the lead wires directly to the contacts on the insulating cylinder. Do not place the shield between the wire and the contact (see Figure 5-10).
- Carefully connect all lead wires. Use the specified tightening torque and ensure all bolt connections are secure and reliable. Use the shields provided with the OLTC to shield the terminals.
- Avoid damaging the insulation on the lead wires when connecting the Tap Selector and the Diverter Switch.
- Securely fasten the OLTC, MDU, and all components when working on the equipment. Prevent accidental tipping or falling of any components.
- Ensure the OLTC is positioned vertically within the support structure (maximum 2° vertical deviation). Install the OLTC on the support structure such that it can be raised 5-20mm to reach its final position after the transformer bell tank is installed.



Tap Selector wiring



Diverter Switch wiring



Cross-section view

Figure 5-10 Connecting Odd and Even Lead Wires between the Tap Selector and Oil Compartment

5.2.4 Installing the OLTC into the Bell-Type Transformer Tank

First, place the OLTC onto a temporary support structure inside the transformer.

Follow these assembly steps:

1. Lift the OLTC onto the support structure and connect it to the regulating winding. Ensure the OLTC is securely positioned and cannot move freely.
2. Use the mounting holes on the support flange to temporarily secure the OLTC to the support structure. If necessary, insert temporary shims between the support flange and the support structure to adjust the OLTC to its final installed height.

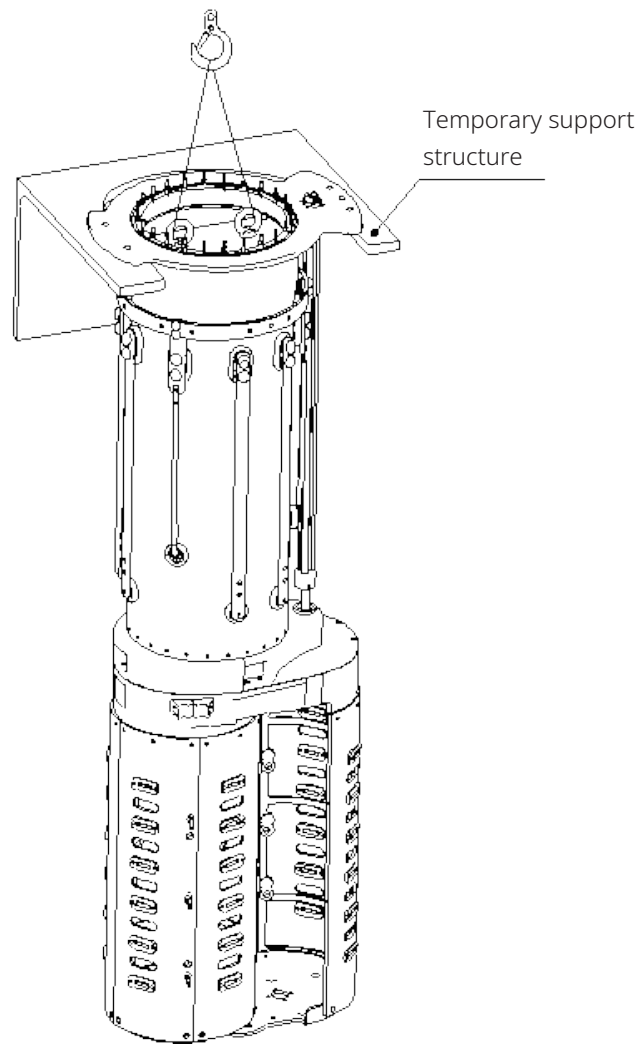


Figure 5-11 Positioning the OLTC on the Support Structure

5.2.5 Assembling the Bell Tank Cover

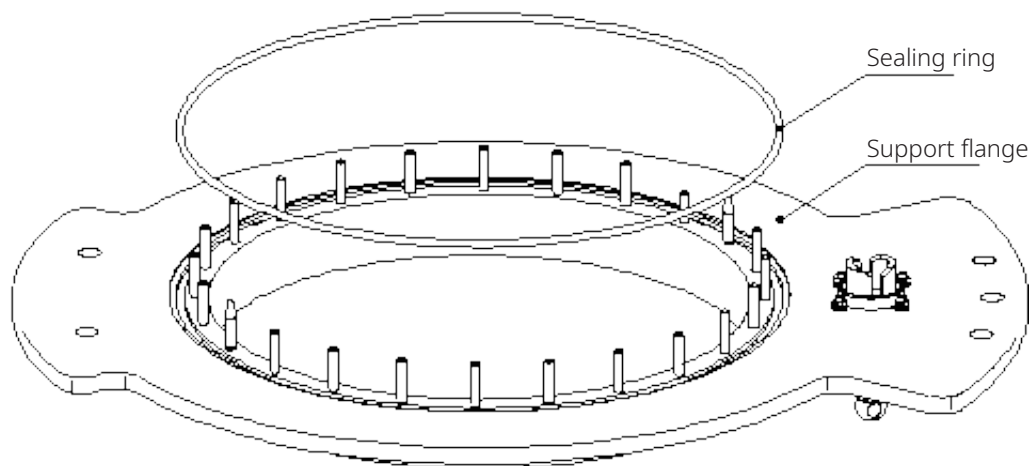


Figure 5-12 Sealing Ring on the Support Flange

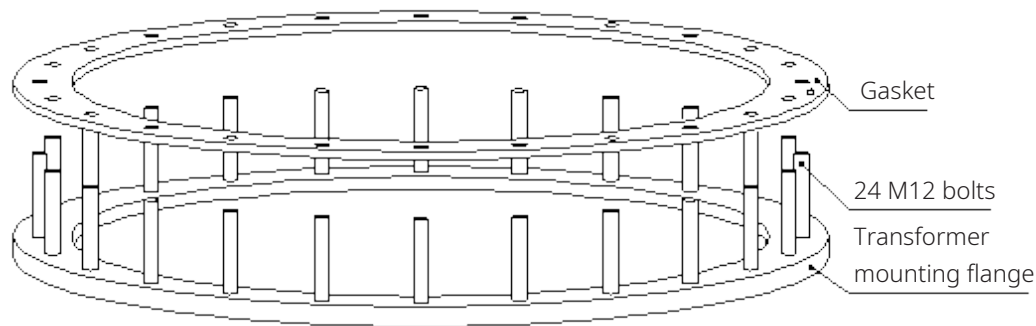


Figure 5-13 Gasket on the Transformer Mounting Flange

Follow these assembly steps:

- Clean the sealing surface on the inside of the Diverter Switch oil compartment support flange and install the sealing ring (see Figure 5-12).
- Clean the sealing surface of the transformer mounting flange and install the gasket (see Figure 5-13).
- 3. Lift the bell tank cover and position it over the transformer. Secure the cover in place.
- 4. Secure the OLTC mounting flange to the transformer mounting flange.
 - Install the 24 M12 bolts and nuts (with washers, use a 19mm wrench).
- 5. Slowly lift the Diverter Switch oil compartment, ensuring all 24 bolts pass through the mounting flange.
 - Once aligned, install nuts on the bolts in the red marked area of the flange (thirteen M8 nuts with washers, use a 13mm wrench). Tighten the nuts sequentially (see Figure 5-14).
 - Remove any temporary shims that were used..

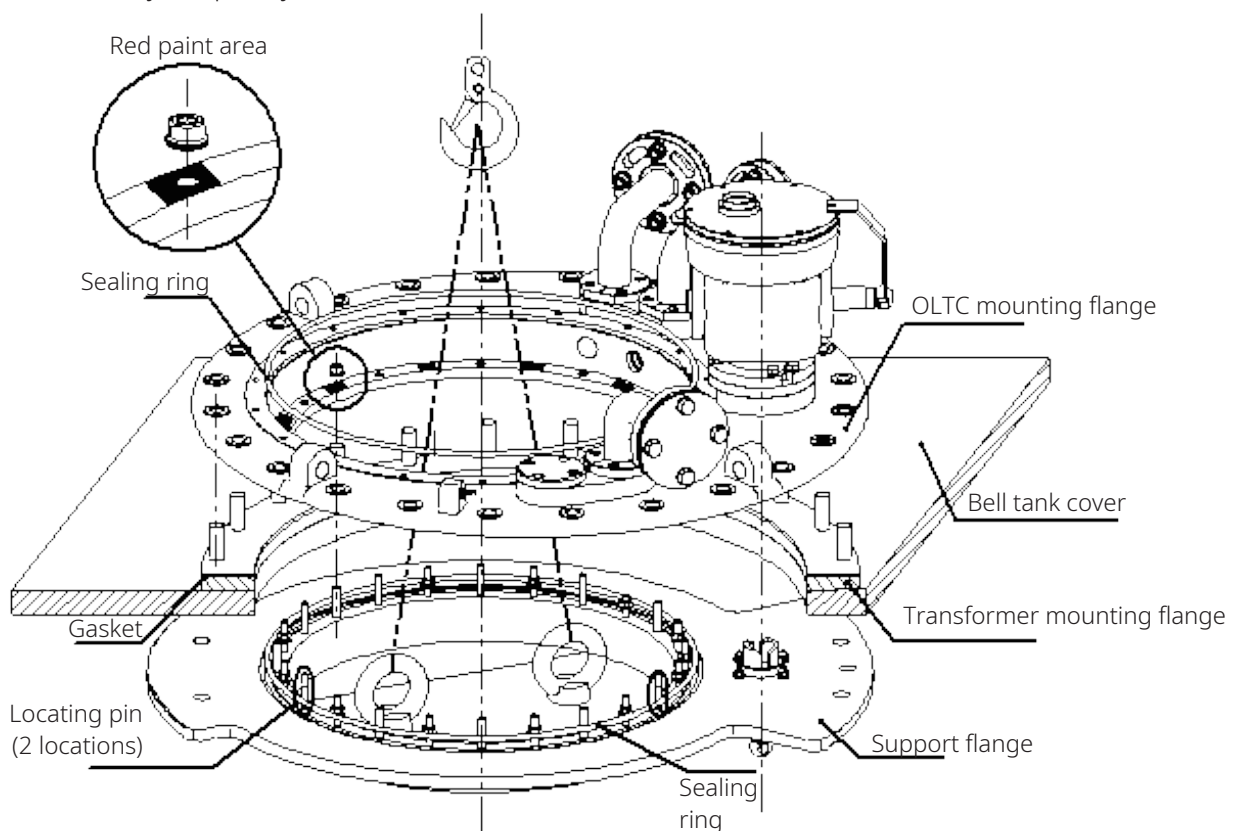


Figure 5-14 Diagram of the OLTC Flanges and Transformer Cover

5.2.6 Installing the Diverter Switch Insert

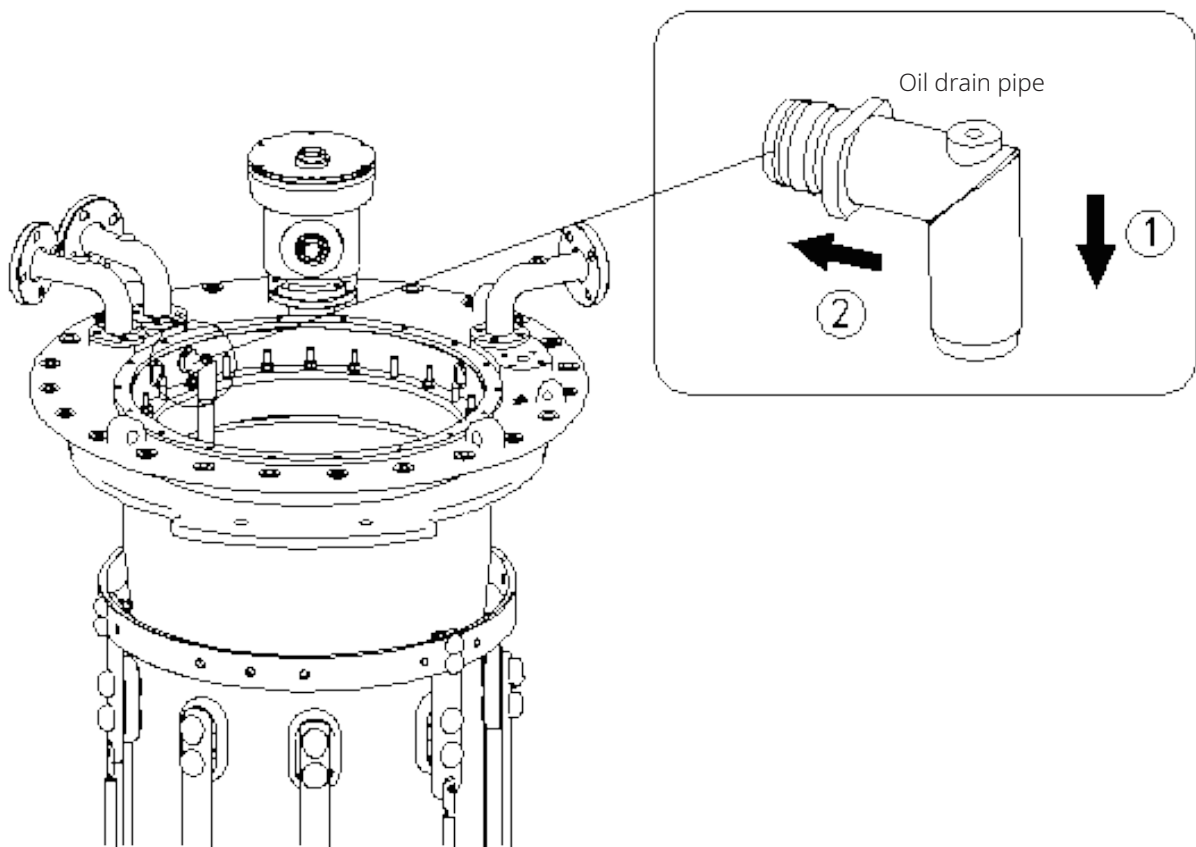


Figure 5-15 Installing the Oil Drain Pipe

1. Install the oil drain pipe.
 - Insert one end of the pipe into the designated hole for the oil drain pipe in the Diverter Switch oil compartment, and insert the other end into the oil drain pipe hole on the OLTC mounting flange.



Caution !

Prevent fasteners from falling into the Diverter Switch oil compartment. Keep all fasteners safely stored for reassembly.

2. Install the Diverter Switch Insert.

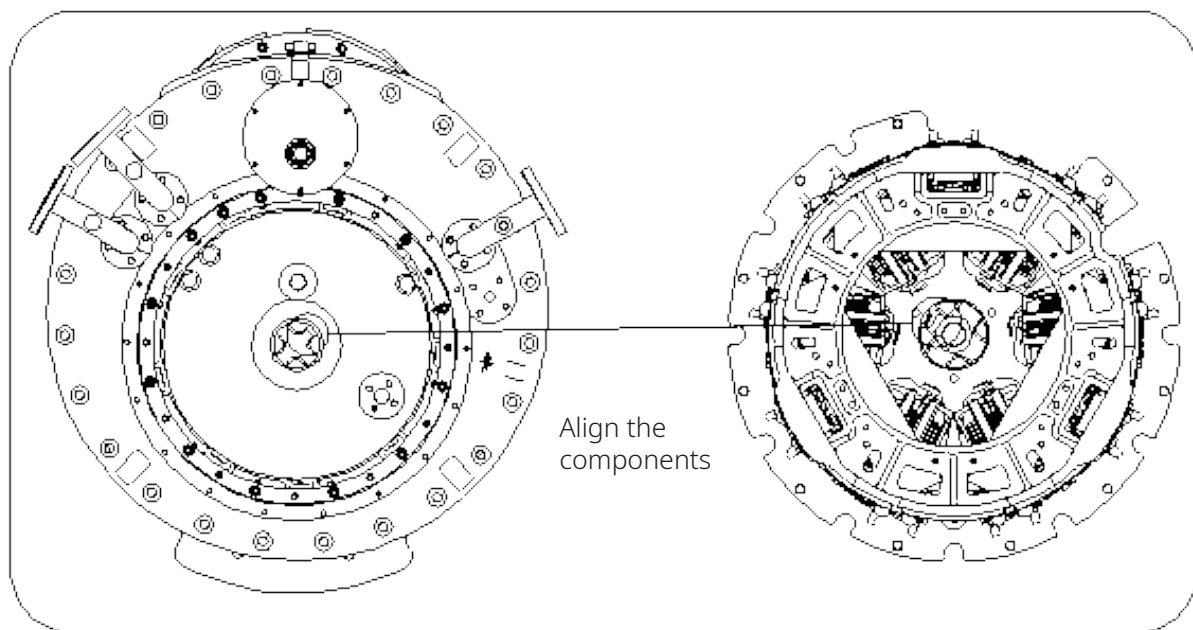
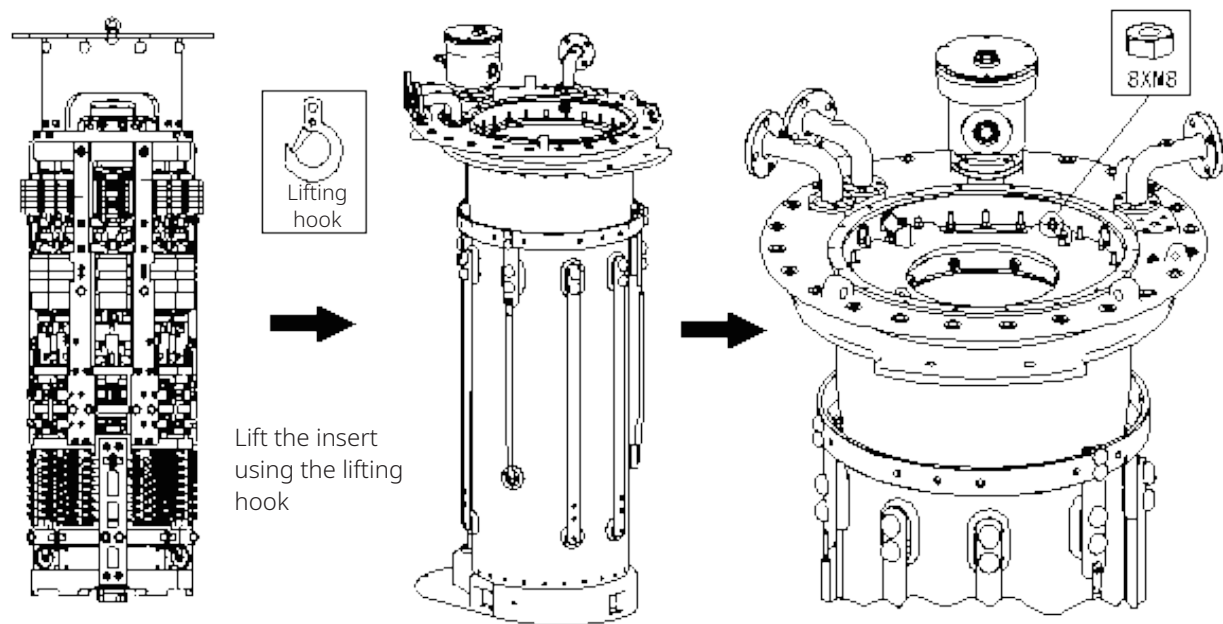


Figure 5-16 Securing Nuts on the Flange Ring and Set Position of the Connecting Parts

Caution !



- After installing the Diverter Switch Insert, inspect the lower surface of the insert's flange ring and the mating surface of the OLTC mounting flange. Ensure they are flush and there is no gap. Otherwise, serious issues may occur, such as failure to switch or tap position errors.
- After installing the Diverter Switch Insert, the upper surface of the insert's flange ring should not exceed the red line.

- Lift the Diverter Switch Insert over the Diverter Switch oil compartment.
 - Ensure the Tap Selector and the bottom connecting part are in their set positions when installing the Diverter Switch Insert.
 - Align the insert with the cross-shaped keyway on the connecting part.
 - After inserting the Diverter Switch Insert into the oil compartment, secure it with nuts in the areas on the flange ring that are not marked with red paint (eight M8 nuts with washers, use a 13mm wrench).
3. Top Cover Installation:
- Inspect the gasket and clean the sealing surfaces.
 - Lift the top cover and position it over the OLTC.
 - Tighten the 18 M8 bolts sequentially (with washers, use a 13mm wrench).

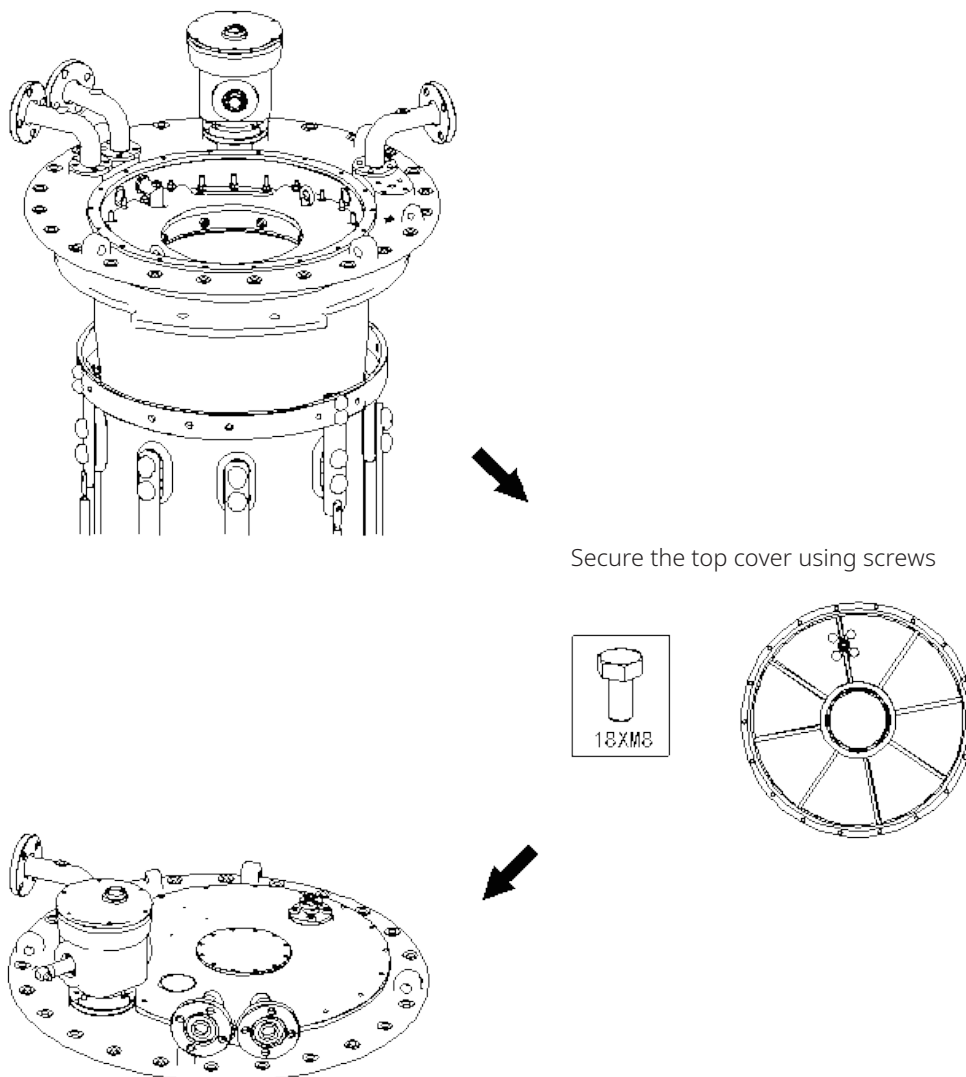


Figure 5-17 Top Cover Installation Steps

**Caution !**

Failure to follow this instruction may damage the equipment.

- Do not damage the O-ring on the oil drain pipe.

5.3 Securing the Regulating Winding Taps to the OLTC Terminals

Caution !

Do not allow the Tap Selector lead wires to exert force on the Tap Selector, causing deformation or damage.

- Evenly distribute the transformer regulating winding lead wires on both sides of the Tap Selector to prevent uneven force on the cage, which can cause twisting and deformation.
- Allow for adequate slack in the lead wires connecting the Tap Selector terminals to the transformer lead wire clamps. The lead wires should be flexible and not excessively short. Avoid applying insulating varnish to this section of the leads to prevent hardening, which can exert force on the insulating rods and cause deformation.
- Connect the lead wire ends to the Tap Selector terminals using an expanded loop configuration to prevent tensile stress on the insulating plates.
- Route the Tap Selector terminal lead wires externally, along the outside of the selector. Do not route them through the interior of the selector.
- □ For bell-type OLTCs, raise the unit 5-20mm after connecting the lead wires. Pay close attention to the tension of the lead wires during this process. It is recommended to install the intermediate flange on the support structure and temporarily raise it to the final installed height before connecting the lead wires (as in the actual installed configuration). After connecting the leads, remove the temporary shims and check the tension of the leads and ensure the OLTC is not subjected to any external forces.



Installation Notice !

Improper installation can damage the OLTC.

- For equipment rated 35kV and above, install the bolt shields provided during installation.
- Use bolts of appropriate length.



The Tap Selector terminals have through-holes for M10 bolts to facilitate connection and securing of the transformer regulating winding taps. The provided shields cover the M10 bolts (see Figure 5-18).

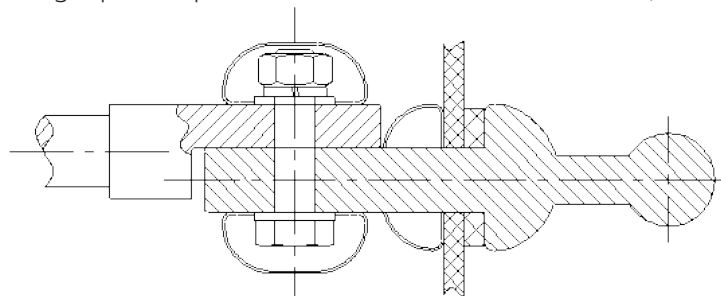


Figure 5-18 Cross-Sectional View of the Regulating Winding Tap Connection to the OLTC Terminal

6. Transformer Turns Ratio Test and DC Winding Resistance Measurement

Important Notes !

- It is recommended to perform the turns ratio test and DC winding resistance measurement before the drying process.
- Operating the OLTC without oil can cause damage.
- Do not operate the OLTC for more than 10 cycles without oil prior to drying.
- Ensure the Tap Selector is fully submerged in oil and the Diverter Switch oil compartment is filled with oil before operating the OLTC for the first time after the transformer drying process.
- Incomplete tap-changing operations can damage the OLTC.



6.1 Turns Ratio Test (Bell-type OLTC Requires Temporary Installation of the Mounting Flange)



Important Note !

- For bell-type OLTCs, temporarily install the mounting flange, Diverter Switch, and top cover for the turns ratio test.

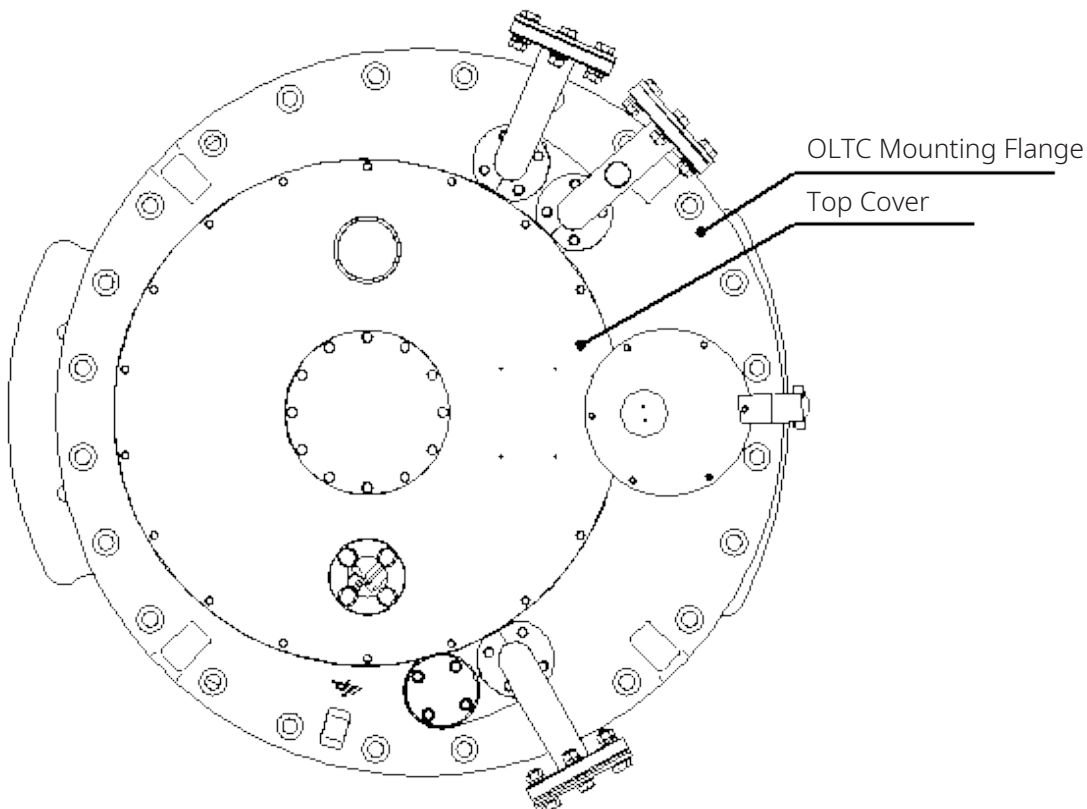


Figure 6-1 Location of Top Components on the OLTC

1. Rotate the drive shaft of the OLTC's spur gear box using a short pipe with an inner diameter of 25mm (insert a 12mm diameter bolt into the pipe) and a handwheel or crank (Figure 6-2).
2. For ganged OLTCs (two or three units), connect all mounting flanges using the horizontal shafts.
3. Each tap change operation will produce an audible click from the Diverter Switch. After the click, continue rotating the shaft in the same direction for two and a half turns to complete the tap change.
4. Observe the tap position through the inspection window on the top cover (Figure 6-2) during operation to avoid exceeding the OLTC's end positions.

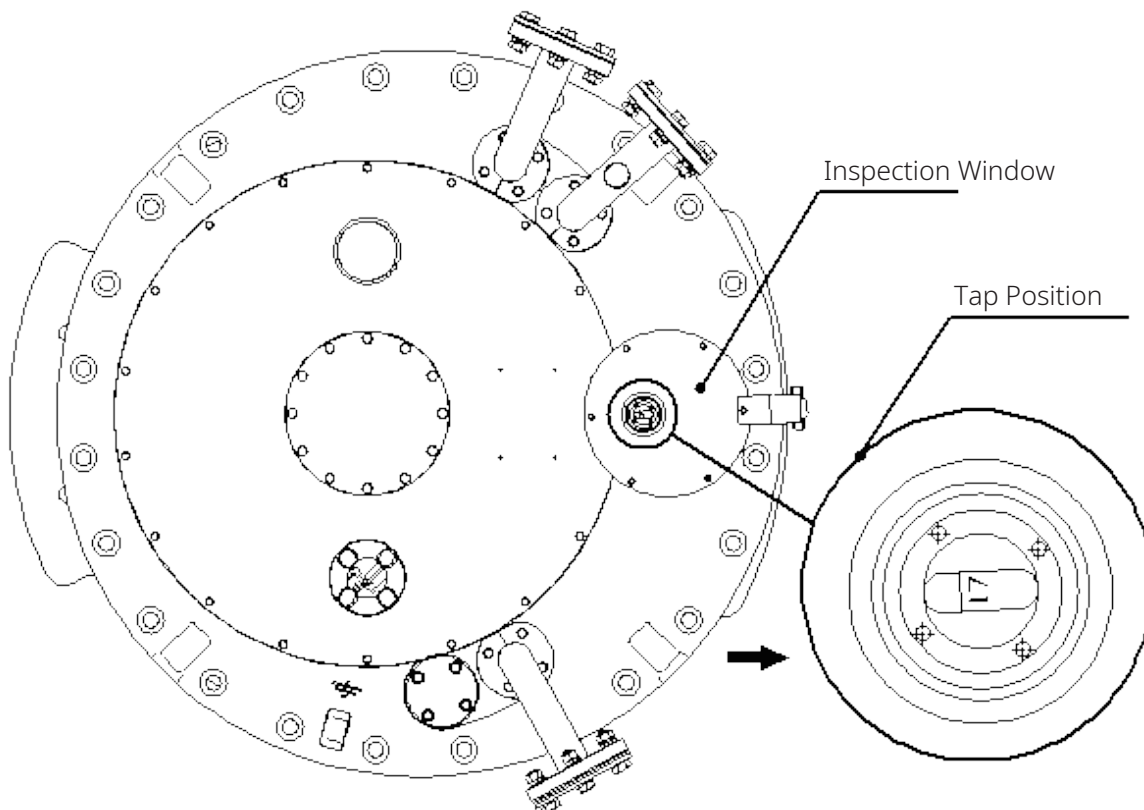


Figure 6-3 Inspection Window on the OLTC

6.2 DC Winding Resistance Measurement

Note that the OLTC is operated without oil during the DC winding resistance measurement. Minimize OLTC operations during this procedure.

After completing the turns ratio test and DC winding resistance measurement, return the OLTC to its designated position.

7. Drying and Oil Filling

To ensure optimal insulation performance, the OLTC must be thoroughly dried according to the following instructions (vacuum drying or vapor-phase drying).

7.1 Drying Process

The Diverter Switch Insert should be dried at a temperature not exceeding 90°C for a maximum duration of 24 hours. If necessary, remove the Diverter Switch Insert (refer to Section 5.5.1 for removal instructions) and dry it separately.

7.1.1 Vacuum Drying

Vacuum Drying Notice !



Improper operation can damage the OLTC.

- Do not place the top cover, gear box, or OLTC accessories in the drying oven with the transformer.
- Keep the OLTC top cover closed during the transformer tank drying process.

Prevent any components from falling into the Diverter Switch oil compartment during disassembly and reassembly. Inspect all components for completeness.



Explosion Hazard !

The presence of flammable materials or ignition sources can result in an explosion, causing severe damage to equipment and potentially fatal or serious injury to personnel.

- Ensure there are no open flames, heat sources, or potential sources of ignition (e.g., static electricity) within the immediate vicinity of the OLTC or within 50 meters of the transformer.

7.1.1.1 Vacuum Drying in a Drying Oven

1. Remove the top cover.

2. Drying Procedure:

- Heat the OLTC in air at normal atmospheric pressure at a rate of approximately 10°C per hour until reaching a final temperature of 110°C.
- Maintain the OLTC at 110°C for 20 hours in circulating hot air.
- Continue drying under vacuum for 50 hours, maintaining the OLTC temperature between 105°C and 125°C. The maximum residual pressure should be 133 Pa.

3. Reassemble the OLTC.

7.1.1.2 Vacuum Drying in the Transformer Tank

1. Disconnect the elbow pipes E2 and R from the OLTC mounting flange (see Figure 7-1).

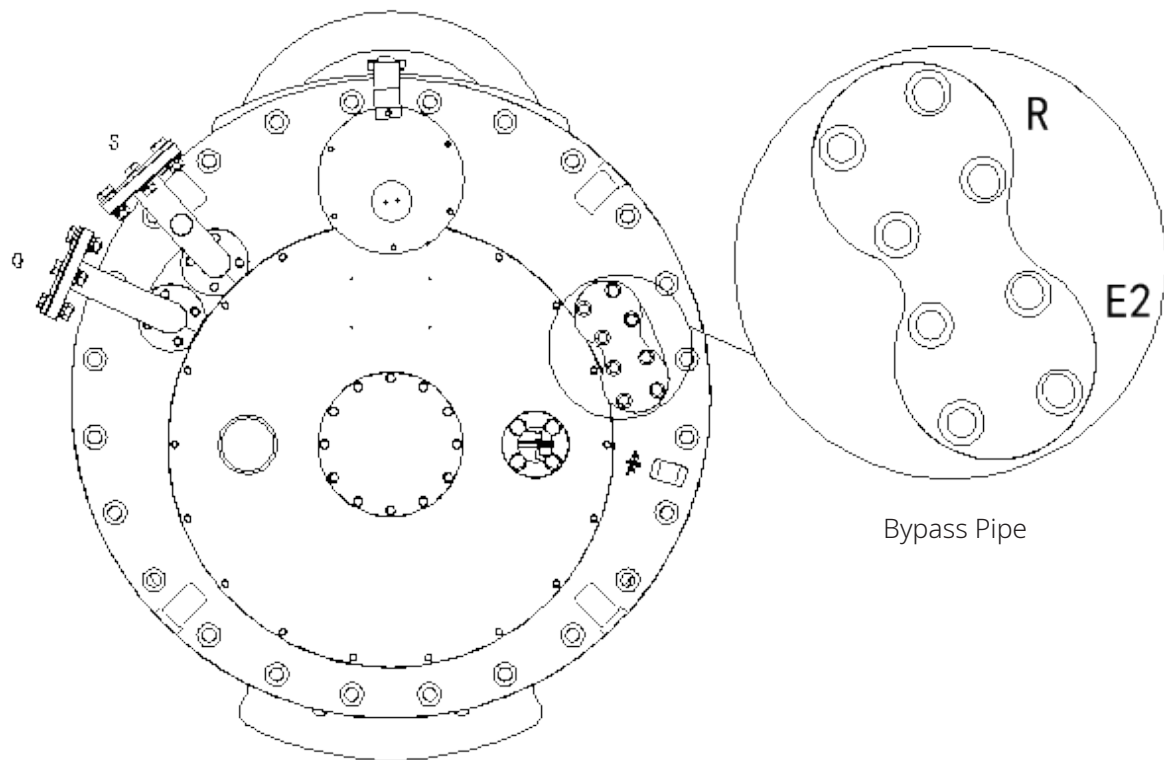


Figure 7-1 Location of Elbow Pipes on the OLTC Top Cover

2. Connect E2 and Q using a bypass pipe with an inner diameter of at least 25mm.
3. Drying Procedure:
 - Heat the OLTC in air at normal atmospheric pressure at a rate of approximately 10°C per hour until reaching a final temperature of 110°C.
 - Maintain the OLTC at 110°C for 20 hours in circulating hot air.
 - Continue drying under vacuum for 50 hours, maintaining the OLTC temperature between 105°C and 125°C. The maximum residual pressure should be 133 Pa.
4. Reassemble the OLTC.

7.1.2 Vapor-Phase Drying

Vapor-Phase Drying Notice !

Improper operation can damage the OLTC.



- Do not place the top cover, gear box, or OLTC accessories in the drying oven with the transformer.
- Prevent any components from falling into the Diverter Switch oil compartment during disassembly and reassembly. Inspect all components for completeness.

Operational Points !

- Use the special wrench to loosen the oil drain bolt by turning it counterclockwise. Do not loosen the bolt more than 10 turns.
- Loosen the oil drain bolt located between the bottom of the Diverter Switch oil compartment and the Tap Selector mechanism by turning it counterclockwise (See Figure 7-2).
- Use the special tool to loosen the bolt from inside the Diverter Switch oil compartment.
- Tighten the oil drain bolt after the drying process.

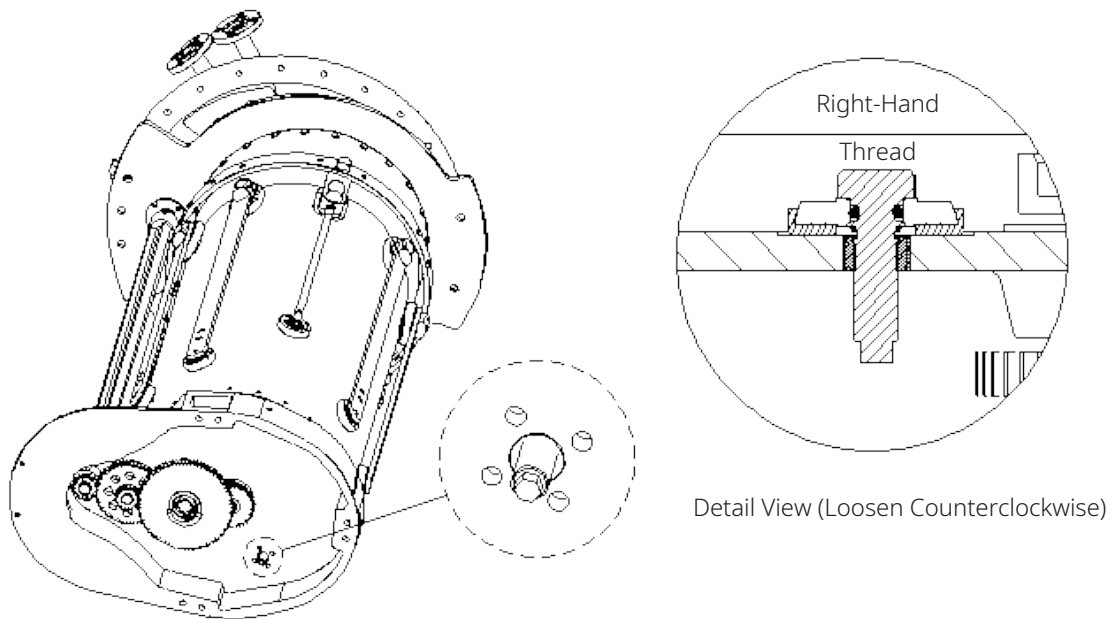


Figure 7-2 Oil Drain Bolt (Schematic)

7.1.2.1 Vapor-Phase Drying in a Drying Oven

**Operational Points !**

- Open the oil drain bolt before placing the OLTC in the drying oven.
- Close the oil drain bolt after the drying process.

1. Remove the top cover.
2. Open the oil drain bolt.
3. Drying Procedure:
 - Inject kerosene vapor at approximately 90°C and maintain a constant temperature for 3-4 hours.
 - Increase the kerosene vapor temperature at a rate of 10°C per hour until reaching the specified final temperature. Ensure the OLTC temperature does not exceed 125°C.
 - Continue drying under vacuum, maintaining the OLTC temperature between 105°C and 125°C. The maximum residual pressure should be 133 Pa. The drying duration should be the same as the transformer drying duration, with a minimum of 50 hours.
4. Reassemble the OLTC.

7.1.2.2 Vapor-Phase Drying in the Transformer Tank

If the transformer does not have a manhole, loosen the oil drain bolt at the bottom of the Diverter Switch oil compartment using the special wrench from inside the oil compartment to allow condensate to drain.

1. Disconnect elbow pipes E2 and R from the OLTC mounting flange (see Figure 7-1).
2. Connect E2 and R using a bypass pipe with an inner diameter of at least 25mm.
3. Remove the top cover.
4. Remove the Diverter Switch Insert (refer to Section 5.2.1 for removal instructions).
5. Open the oil drain bolt.
6. Reinstall the Diverter Switch Insert (refer to Section 5.2.6 for installation instructions).
7. Drying Procedure:
 - Inject kerosene vapor at approximately 90°C and maintain a constant temperature for 3-4 hours.
 - Increase the kerosene vapor temperature at a rate of 10°C per hour until reaching the specified final temperature. Ensure the OLTC temperature does not exceed 125°C.
 - Continue drying under vacuum, maintaining the OLTC temperature between 105°C and 125°C.

The maximum residual pressure should be 133 Pa. The drying duration should be the same as the transformer drying duration, with a minimum of 50 hours.
8. Reassemble the OLTC.

7.2 Oil Filling

Operational Points !



- Avoid damaging the sealing surface of the OLTC mounting flange during disassembly and reassembly.
- Ensure the Tap Selector is fully submerged in transformer oil and the Diverter Switch oil compartment is filled with oil before operating the OLTC for the first time after the transformer drying process.
- If the OLTC is dried inside the transformer tank, loosen the oil drain bolt. Tighten the oil drain bolt after drying.

After completing the drying process, refill the Diverter Switch oil compartment (including the Diverter Switch Insert) with oil as quickly as possible to minimize moisture ingress.

Procedure:

Use only new transformer oil that meets the required standards for filling the Diverter Switch oil compartment and its conservator.

Fill the OLTC and the transformer with new oil simultaneously under vacuum..

1. Reinstall the top cover.
 - Ensure the O-ring is properly positioned.
 - Tighten the 24 M10 bolts.

2. Install the bypass pipe.
 - Connect the bypass pipe between fittings E2 and Q.
 - Ensure the OLTC and transformer are simultaneously under vacuum.
3. Fill the OLTC and transformer with new transformer oil.

8. Installation of the OLTC Protective Device and Drive Unit Assembly

8.1 Gas Relay Installation

Install the gas relay according to the instructions provided in the relay's manual.

8.2 Pressure Relief Device Operation

Before operating the OLTC, remove the red crossbar from the top of the Pressure Relief Device.

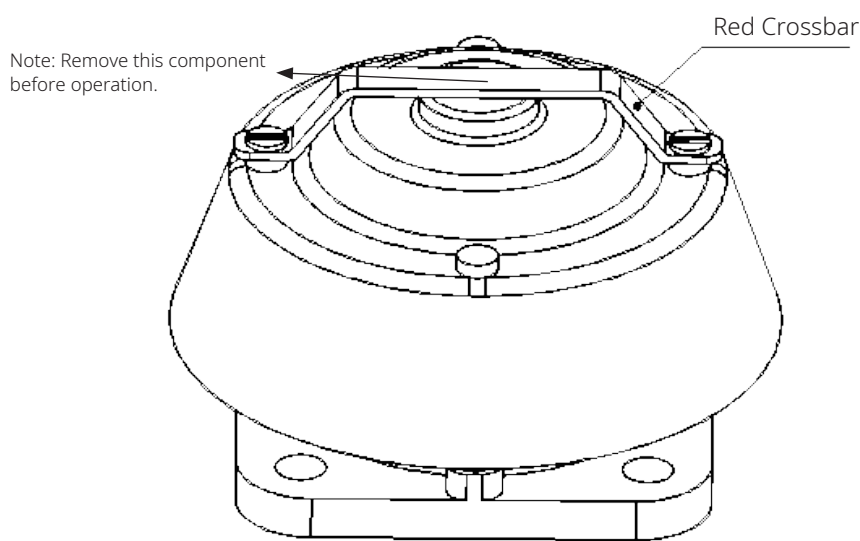


Figure 8-1 Red Crossbar on the Pressure Relief Device

8.3 Motor-Drive Unit (MDU) Installation

The MDU provides the motive force for tap-changing operations and can be operated electrically or manually.

MDU Installation Notes:



- Connect the MDU to the OLTC only when both are in their designated positions.
- Mount the MDU vertically on the transformer tank wall, ensuring proper alignment.
- Ensure the MDU mounting plate is flat to prevent twisting or deformation of the MDU, which could impact its operation.
- Refer to the Huaming MDU instruction manual for detailed installation instructions.

8.4 Adjusting the Spur Gear Box

To adjust the spur gear box, follow these steps:

1. Loosen the six bolts that secure the clamping plates beneath the spur gear box. It is not necessary to remove the bolts completely.

2. Slightly pry the three clamping plates outwards. This will allow you to rotate the spur gear box and adjust the output shaft to face the MDU.
3. Return the clamping plates to their original position and tighten the six bolts. Ensure the spur gear box is securely mounted and there is no movement (see Figure 8-2)..

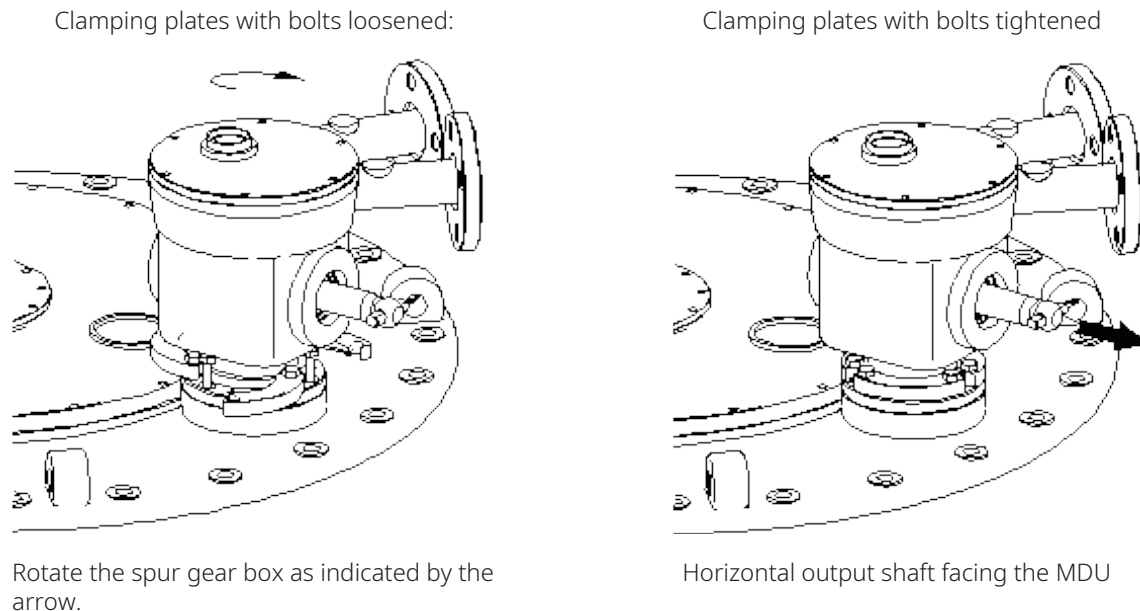


Figure 8-2 Adjusting and Securing the Spur Gear Box

8.5 Bevel Gear Box Installation

Secure the bevel gear box to the transformer cover bracket using two bolts (18mm diameter holes).



Note:

- Ensure the factory number on the bevel gear box nameplate matches the factory number on the OLTC nameplate.

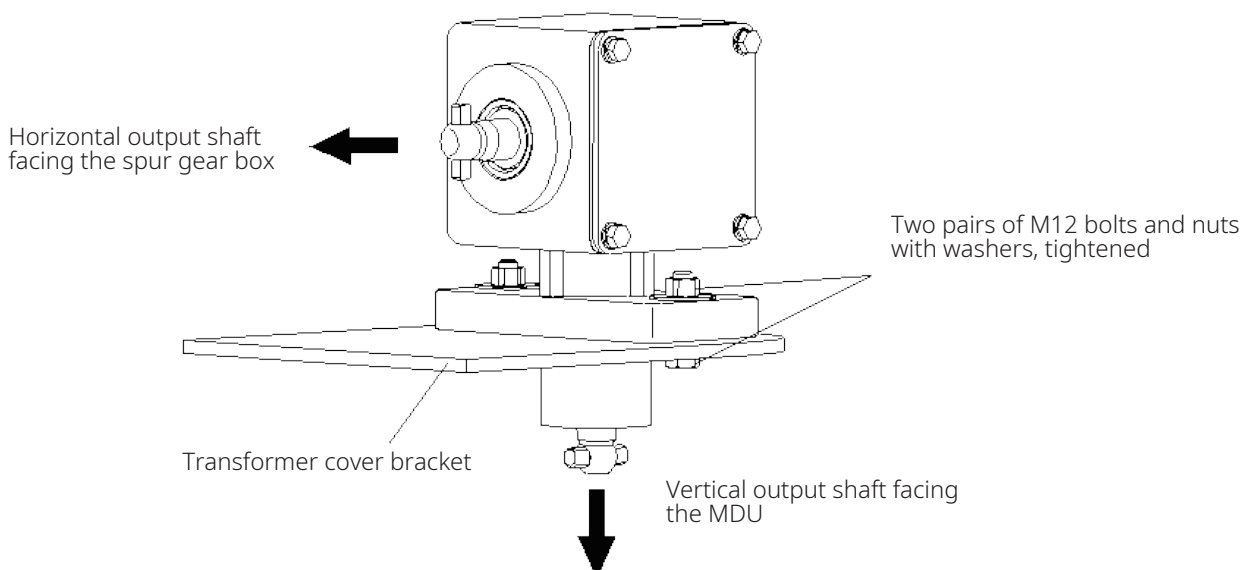


Figure 8-3 Bevel Gear Box Installation Diagram

8.6 Horizontal and Vertical Drive Shaft Installation

Refer to the Huaming MDU instruction manual for instructions on connecting the drive shafts, MDU, and OLTC.

Figure 8-4 illustrates the installation of the vertical drive shaft coupling half to the lower end of the bevel gear box. Install the horizontal drive shaft coupling half in the same manner.

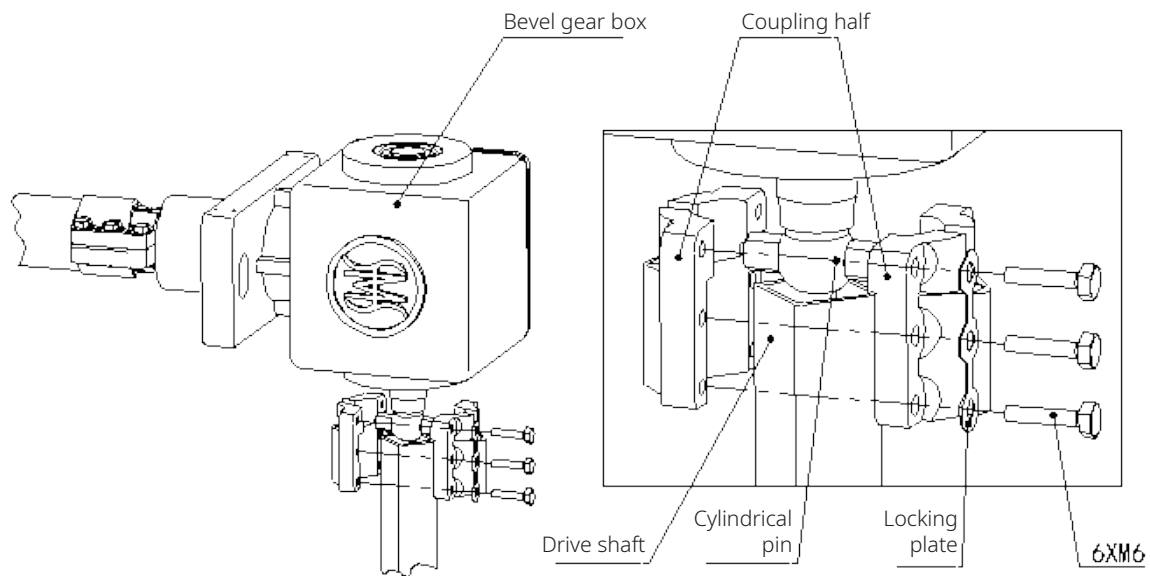


Figure 8-4 Installation of the Bevel Gear Box Lower Connection

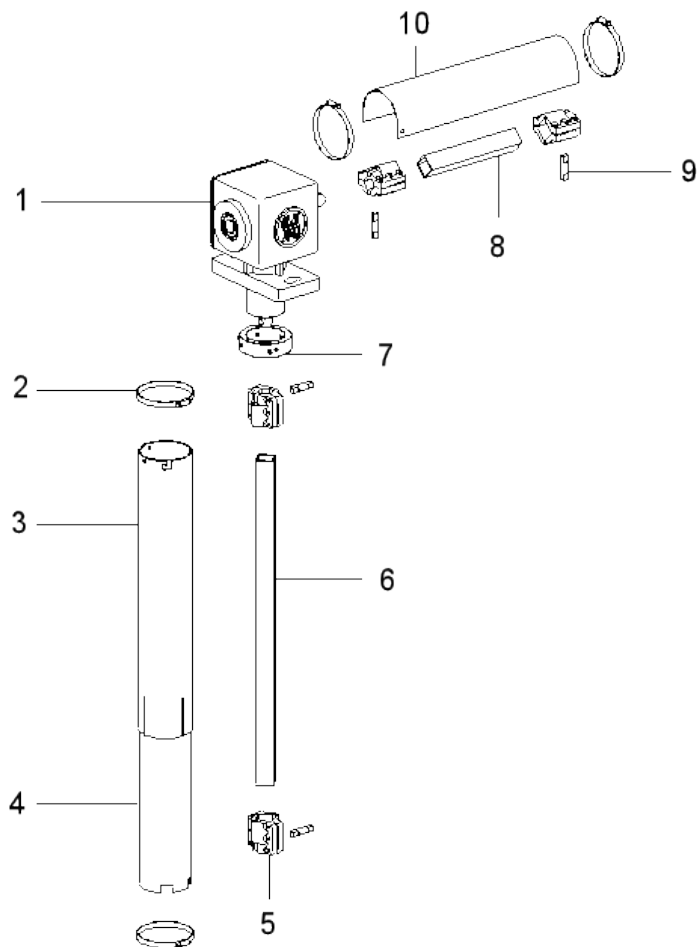
Drive Shaft Installation Notes :



- For horizontal and vertical drive shafts exceeding 2 meters in length, install an intermediate support gear box. Refer to the above instructions for the installation of specially designed bevel gear boxes, steering gear boxes, and intermediate support gear boxes for vertical or horizontal drive shafts.

8.7 Drive Shaft Guard Installation

8.7.1 Guard Installation Diagram



1. Bevel gear box	2. Clamp
3. Vertical guard 1	4. Vertical guard 2
5. Coupling	6. Square tube 1
7. Bushing	8. Square tube 2
9. Cotter pin	10. Horizontal guard

Figure 8-5 Drive Shaft Guard Installation (Schematic)

8.7.2 Bushing Installation



Installation Notice !

- The cylindrical pin should not protrude from the inside of the bushing, as this can interfere with installation.

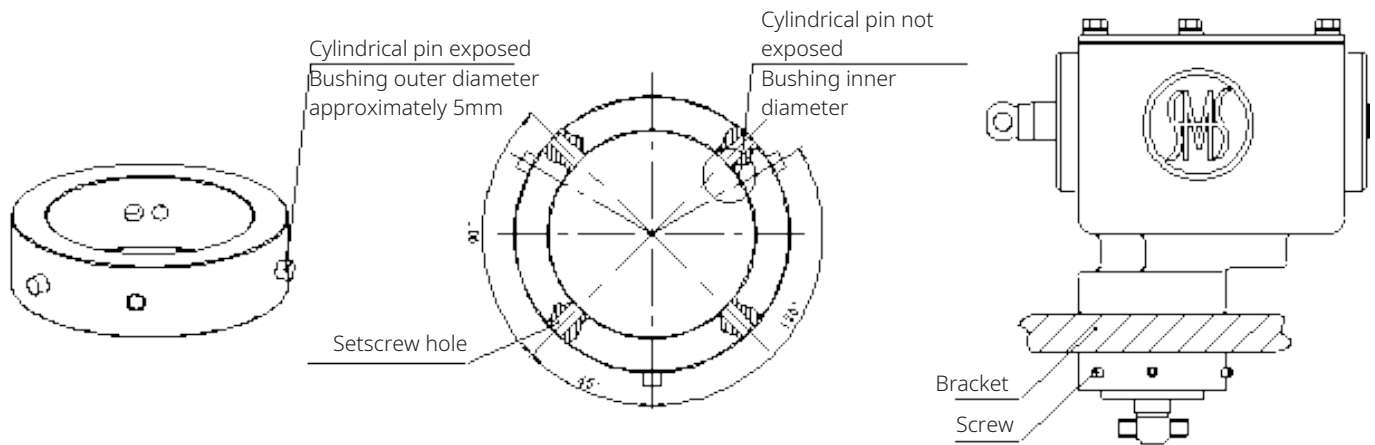


Figure 8-6 Installation Correcte de la Douille

8.8 Installation of Three Single-Phase OLTCs and I+II OLTCs

Note !



The following precautions apply to the connection of three single-phase OLTCs and I+II OLTCs:

- When three single-phase OLTCs or an I+II OLTC configuration are installed on the same transformer, adjusting the gear boxes may cause the Diverter Switches to operate. Therefore, after adjusting the gear boxes, verify that all Diverter Switches are in the same position. Manually operate the system and confirm that all Diverter Switches operate simultaneously.

8.9 Calibration of the OLTC Drive System

Before electrically operating the MDU, manually operate the OLTC through one complete cycle to ensure proper alignment and function.

The time interval between Diverter Switch operation and the completion of MDU movement should be consistent for both rotational directions.

To ensure reliable operation, calibrate the drive system after disconnecting and reconnecting any horizontal or vertical shafts.

8.9.1 Calibration Procedure:

1. Manually rotate the drive shaft from position 1 to N using the crank handle. Once you hear the Diverter Switch operate (audible click), continue rotating the handle and count the number of rotations (m) until the center mark of the gray area on the MDU's tap position indicator aligns with the arrow.
2. Rotate the handle in the opposite direction (N to 1) back to the original set position, counting the number of rotations (k) using the same method.
3. If $m = k$, the connection is correct. If $m \neq k$ and the difference ($|m - k|$) is greater than 1, rotate the handle $1/2(|m - k|)$ turns in the direction with the greater number of rotations. Then, disconnect and reconnect the vertical drive shaft to the MDU.
4. Repeat the above steps to check the rotation difference between the MDU and the OLTC connection until the number of rotations for both directions is approximately equal.

8.9.2 Example :

Calibrating an OLTC with Tap Selector 10193W:

- Rotate from position 10 (set position) to position 11. The number of handle rotations (m) is 5.
- Rotate from position 11 back to position 10 (original set position). The number of handle rotations (k) is 3.
- The difference in handle rotations is: $|m - k| = |5 - 3| = 2$ rotations.
- The adjustment rotation is $1/2(m - k) = 1/2(5 - 3) = 1$ rotation.
- Disconnect the vertical drive shaft from the MDU. Rotate the handle one turn in the direction from position 10 to position 11. Reconnect the vertical drive shaft.
- Verify that the rotation difference is balanced for both rotational directions.

Calibration Summary:

1. Record the number of rotations for both directions (m and k).
2. Disconnect the coupling and rotate the handle $1/2(|m - k|)$ turns in the direction with the greater number of rotations.
3. Reconnect the coupling and repeat the calibration until the rotation difference ($|m - k|$) is less than 1.

9. Factory Acceptance Tests (FAT) and Pre-Test Preparations

9.1 Pre-Test Preparations

9.1.1 Complete Degassing of the OLTC

Before initial operation, de-gas the OLTC head and the oil drain pipe connected to fitting S using the bleeder plugs.

1. Degassing the OLTC Head:

- Remove the M30 bolt from the bleeder valve E1 (use a 36mm wrench).
- Use a screwdriver to pry up the valve stem and allow air to escape from the OLTC head until oil flows out (see Figure 9-1).
- Tighten the bolt cap on bleeder valve E1.

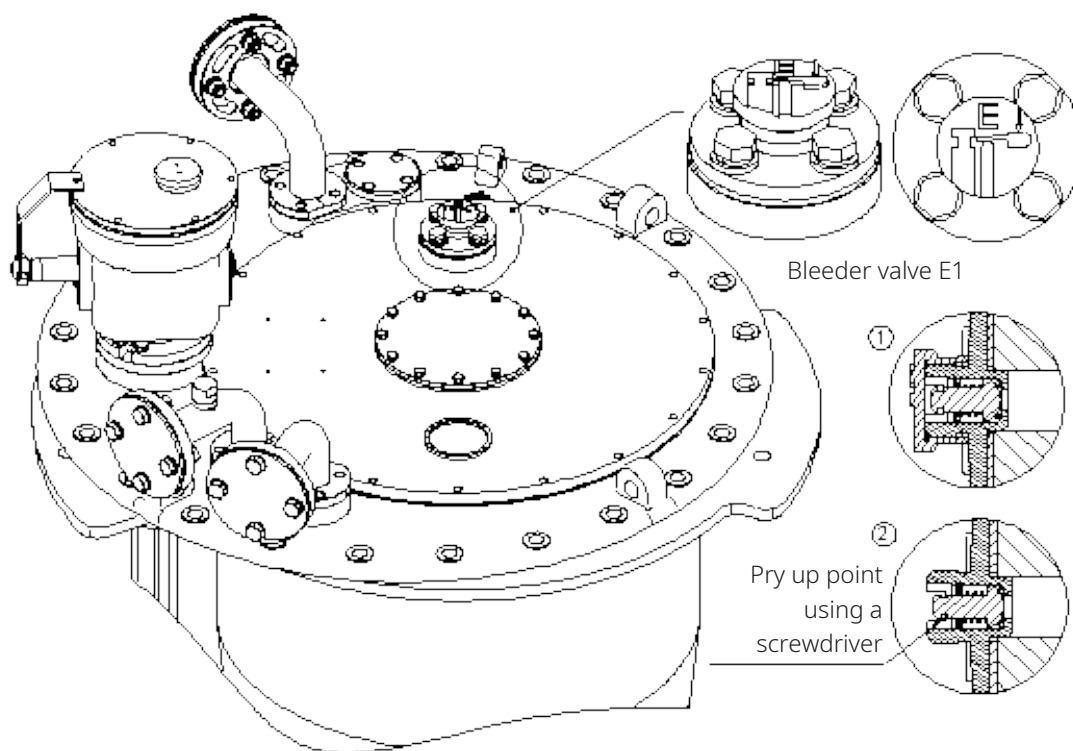


Figure 9-1 Degassing Procedure for Bleeder Valve

2. Degassing the Oil Drain Pipe at Fitting S:

- Remove the M16 bolt cap from fitting S (use a 24mm wrench, see Figure 9-2).
- Open the M6 bleeder plug with the groove to release air from the oil drain pipe.
- Close the bleeder plug.
- Replace the bolt cap on fitting S.

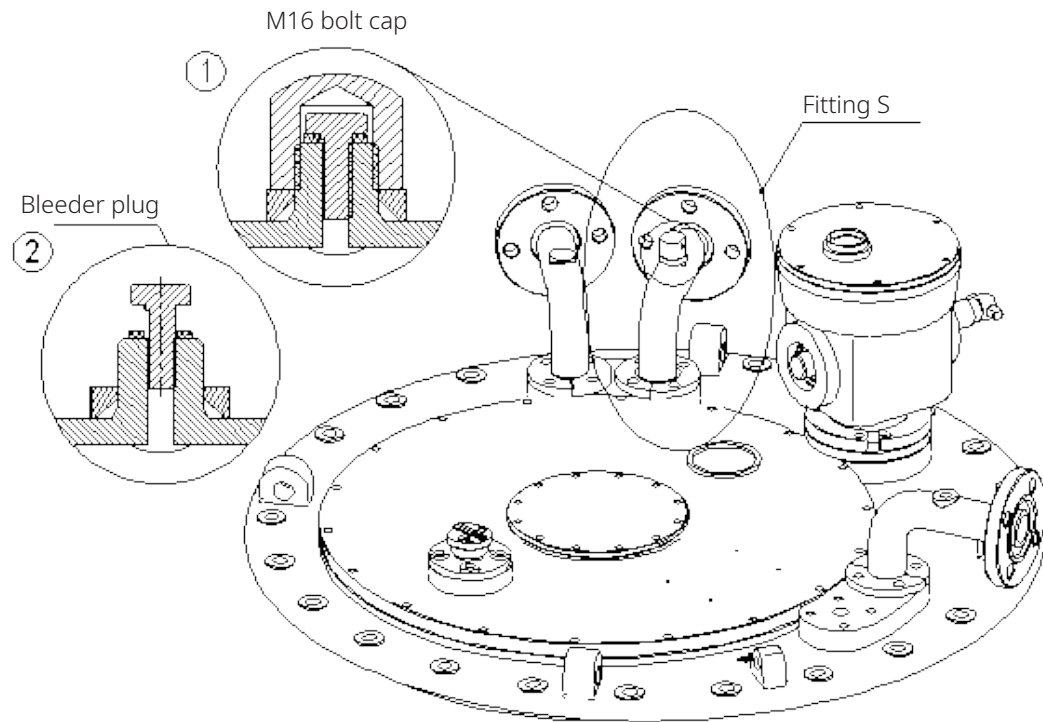


Figure 9-2

9.1.2 Grounding

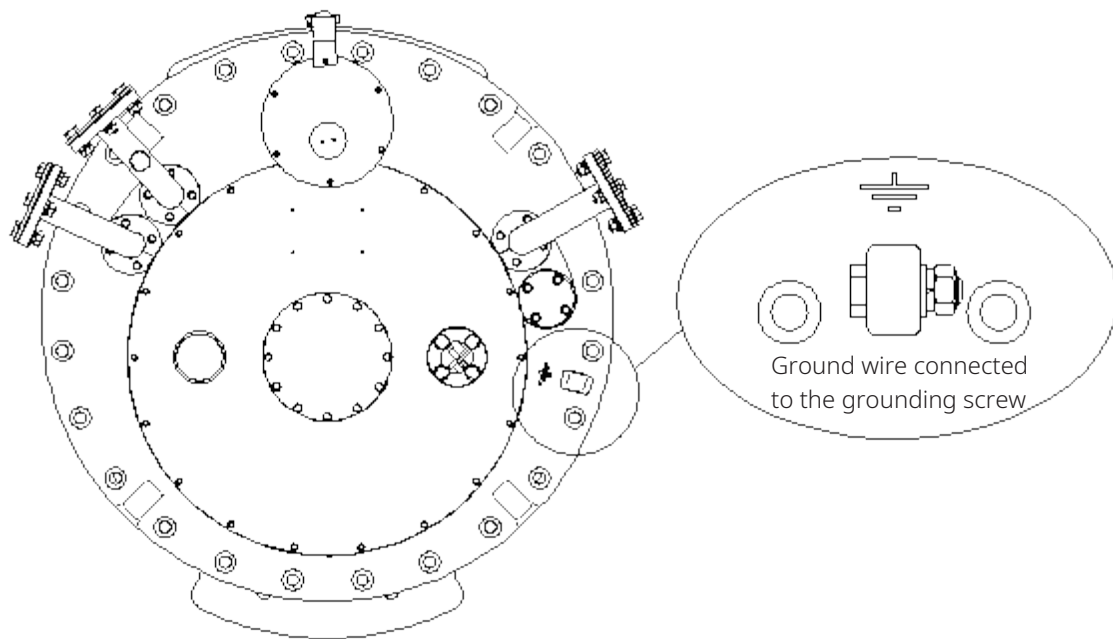


Figure 9-3 Connecting the Ground Wire to the Grounding Screw

1. Connect the grounding screw on the OLTC mounting flange to the transformer cover using the provided M12 bolt and nut (19mm wrench, see Figure 9-3).
2. Connect the M12 grounding screw on the MDU protective enclosure to the transformer tank (19mm wrench).

9.2 Tests at the Transformer Manufacturer's Facility

9.2.1 Functional Test

Before energizing the transformer, perform a functional test to verify the mechanical operation of the OLTC and MDU.

Caution !

Observe the following during the functional test:



- Ensure the Tap Selector is fully submerged in oil and the Diverter Switch oil compartment is filled with oil before operating the OLTC for the first time after the transformer drying process. Do not operate the OLTC without oil.
- Verify that the Tap Selector is fully immersed in oil and the Diverter Switch oil compartment is filled with oil.
- Operate the OLTC through its entire range during the functional test (one complete cycle).
- Ensure the position indicators on the MDU and the OLTC (view through the inspection window on the top cover) display the same tap position at each step.

Installation Warning !

Improper connection between the OLTC and the MDU can cause damage to the OLTC.



- Do not operate the OLTC if there is an improper connection between the OLTC and the MDU.
- Ensure the position indicators on the MDU and OLTC display the same tap position.

9.2.2 Transformer Electrical Tests

Flammable and Explosive Gases !

Flammable and explosive gases can accumulate within the OLTC, posing a risk of explosion, severe equipment damage, and potentially fatal or serious injury.



- Explosive gases may be present under the OLTC top cover, within the connection system, inside the conservator, and at the outlet of the air breather. Ejected components and hot oil spills can cause severe or fatal injuries.

Explosion Hazard !

The presence of flammable materials or ignition sources can result in an explosion, causing severe equipment damage and potentially fatal or serious injury.



- Ensure there are no open flames, heat sources, or potential sources of ignition (e.g., static electricity) within the immediate vicinity of the OLTC or within 50 meters of the transformer. All OLTC tests must be carried out by qualified personnel according to established test procedures.

10. Transformer Transportation and On-Site Commissioning

10.1 Removing the MDU for Transportation

1. Set the MDU to its designated position and disconnect the coupling.
2. Remove the MDU.
3. Do not operate the MDU if it is not connected to the OLTC. Refer to the Huaming MDU instruction manual for re-installation procedures.

10.2 Transportation with Oil Filling and No Conservator

**Note !**

- During a storage and transportation period of 3-4 weeks without a conservator, reduce the oil volume in the OLTC by approximately 5 liters.

If the transformer is filled with oil during storage and transportation but is not equipped with a conservator, install a bypass pipe between the Diverter Switch oil compartment and the transformer tank to equalize pressure changes due to oil expansion. Connect the bypass pipe between fittings E2 and R on the OLTC mounting flange.

10.3 Transportation without Oil Filling

**Note !**

- During long-term storage, periodically energize the MDU heater to prevent moisture accumulation.

If the transformer is transported or stored without oil filling, drain all oil from the Diverter Switch oil compartment. Protect the oil compartment interior using a method suitable for transformer preservation, such as nitrogen blanketing.

10.4 On-Site Commissioning

**Caution !**

- Operating the OLTC without oil will cause damage.
- Ensure the Tap Selector is fully submerged in oil and the Diverter Switch oil compartment is filled with oil before energizing the transformer.



Explosion Hazard !

The presence of flammable materials or ignition sources can result in an explosion, causing severe equipment damage and potentially fatal or serious injury to personnel.

- Ensure there are no open flames, heat sources, or potential sources of ignition (e.g., static electricity) within the immediate vicinity of the OLTC or within 50 meters of the transformer.

10.4.1 On-Site Procedures :

1. Before initial energization, verify that the transformer oil meets the following requirements: breakdown voltage (Ud) > 40kV/2.5mm (minimum) and water content < 30µL/L (maximum).
2. De-gas the OLTC as described in Section 9.1.1.
3. Perform a functional test as described in Section 9.2.
4. Connect the minimum oil level signal contact in the OLTC conservator to the circuit breaker trip circuit.
5. Verify that the gas relay is functioning correctly according to the instructions provided in the Huaming QJ-25 series protective relay manual.
6. Remove the red protection strip from the Pressure Relief Device before energizing the transformer.

Ensure the inrush current transient has subsided before performing any tap-changing operations. Inrush current can be several times higher than the normal operating current and may overload the OLTC during switching. It is recommended to wait at least 30 seconds after energization, allowing the current to stabilize, before operating the OLTC.

Observe all safety precautions and instructions outlined in this manual during functional checks, testing, and commissioning.

After energizing the transformer and allowing the inrush current transient to subside, tap-changing operations can be performed under no-load or load conditions.

11. Monitoring during on-load tap-change operation

11.1 Monitoring



CAUTION !

Visual inspection of on-load tap-changer head cover, protective relay and motor-drive unit should be carried out regularly.

Regulation type	Dielectric strength	Moisture content
At neutral point of winding	$\geq 30 \text{ KV}/2.5 \text{ mm}$	$< 40 \text{ ppm}$
At any points other than neutral point	$\geq 40 \text{ KV}/2.5 \text{ mm}$	$< 30 \text{ ppm}$

11.1.1 In case of transformer overload, the on-load tap-changer must not be operated frequently and must be equipped with an "overcurrent self-closing contact" so that the on-load tap-changer is not switched when the load current is greater than 2 times I_n .

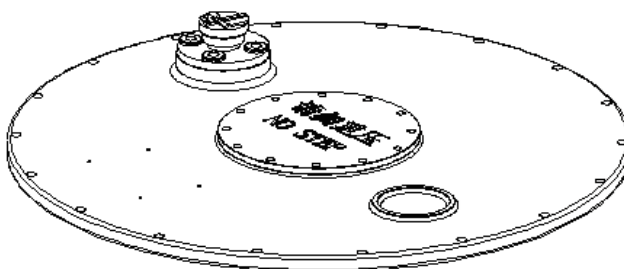
11.1.2 The trip contact of protective relay is set to operate when the oil speed is $1.0 \text{ m/s} \pm 10\%$. This contact should be connected to the tripping circuit of the circuit breaker; When the fault occurs inside the on-load tap-changer, a large amount of gas will be generated, resulting in accelerated oil flow rate, so that the baffle of protective relay is activated, trip contact closes, to cut off the transformer power supply and avoid the expansion of the accident. Once the protective relay is activated, it is strictly prohibited to switch on circuit breaker before lifting out diverter switch insert for check.

11.1.3 Rupture disk is installed on the on-load tap-changer head cover, which will not work during the normal operation of the tap-change. The rupture disk will break only if failure happened inside diverter tap changer oil compartment and the pressure in the oil compartment exceeds $0.3 \pm 20\% \text{ MPa}$ to avoid escalating the accident. During on-load tap-changer installation and maintenance, be careful not to step on or hit the rupture disk with heavy objects.



CAUTION !

No step on rupture disk!



12. Notes for on-load tap-changer Maintenance

Electric shock warning! (Danger for Death or severe injury)

An energized transformer may cause death or severe injury.



- Turn off the high voltage and low voltage terminals of the transformer.
- Lock the transformer to prevent accidental restart.
- Ensure that all components are powered off.
- Ensure that all transformer terminals (earthing wires, earth disconnect switches) are earthed and short-circuited.
- Cover or isolate neighboring live parts.

Electric shock warning! (Danger for Death or severe injury)

Operation of energized on-load tap-changers may cause personnel death or severe injury.



- Disconnect all auxiliary circuits, such as tap changer monitoring devices, pressure relief valves and pressure monitors.
- Ensure that all components are powered off.

Warning! (Danger for Death or severe injury)

Risk of explosion!



- Explosive gas in the diverter tap-changer oil compartment, transformer tank, connection pipes, oil conservator, and the outlet of the dehydrating breather can ignite or explode, resulting in severe injury or death.
- Ensure that there are no open flames, hot surfaces or electrical sparks (e.g. caused by static build-up) in the environment surrounding the transformer.
- Do not operate any electrical devices (for example, impact wrenches have the risk of spark).
- Only use conductive and grounded hoses, pipes and pump-operated equipment which are suitable for flammable liquids.

Check the motor-drive unit!

Condensation in the motor-drive unit can cause damage to the motor of the motor-drive unit.



- Always keep the cabinet of motor-drive unit sealed.
- If the motor-drive unit is out of work for more than 2 weeks, connect and operate the anti-condensation heater in the motor-drive unit. If the condition is not allowed (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the cabinet.

13. Inspection and maintenance

13.1 Regular inspection

Please follow the inspection instruction below:

Interval	Instruction	Details
Once per year	Check the door sealing, cable sleeves, motor-drive unit cabinet and control panel's ventilation device	Visually check. If the sealing parts are aging and broken, and there is oil leakage, replacement is required immediately. The ventilation of the motor drive unit and control panel should be tested. If aging damage occurs, replacement is required immediately.
Once per year	Check the sealing of on-load tap-changer top cover, protective relay and all pipe connections.	Visually check. If the sealing parts are aging and broken, and there is oil leakage, replacement is required immediately.
Once per year	Check if the heater inside the motor-drive unit cabinet works properly.	Visually check. If the electronic components are getting black or yellow, and the pins are broken, please contact Huaming service. Then switch on temperature controller and check its functionality use a thermometer.
Once per year	Check the protective relay.	Visually check. If the electronic components are getting black or yellow and the pins are broken, please contact Huaming service.
Once per year	Check if the silica gel in the breather of on-load tap-changer oil conservator is working properly	Put the humidity indicator card in the oil conservator for at least half an hour, if the humidity is greater than 10%, replace the silica gel, or replace it every other year.
Once per year or every 200,000 operations	Check the oil sample.	Discharge the transformer oil inside pipe for around 5L, then take the oil sample from diverter switch. If the parameter of the oil sample does not meet with the data in Table 12.1.2, it is recommended to filter or replace the transformer oil.

Table 13.1.1 Inspection instruction

Item		Dielectric strength	Water content
Limit values for insulating oil in accordance with IEC 60296			
When commissioning the transformer for the first time		> 60kV/2.5mm	< 15ppm
During operation	Class I tap-changer	> 30kV/2.5mm	< 40ppm
	Class II tap-changer	> 40kV/2.5mm	< 30ppm
After maintenance		> 45kV/2.5mm	< 15ppm
Limit values for natural esters in accordance with IEC 62770			
When commissioning the transformer for the first time		> 60kV/2.5mm	≤ 100ppm
During operation	Class I tap-changer	> 30kV/2.5mm	≤ 200ppm
	Class II tap-changer	> 40kV/2.5mm	≤ 200ppm
After maintenance		> 45kV/2.5mm	≤ 100ppm

Table 13.1.2 Oil sample limit values

13.2 Maintenance

Interval	Instruction
After 100,000 times on-load tap-changer operation (based on the number shown in motor drive-unit's counter)	on-load tap-changer inspection and maintenance. Please contact Huaming's service department.
After 200,000 times on-load tap-changer operation (based on the number shown in motor drive-unit's counter)	Inspection for vacuum interrupters. Please contact Huaming's service department.
After 300,000 times on-load tap-changer operation (based on the number shown in motor-drive unit's counter)	Replace the diverter switch insert. Please contact Huaming's service department.
After 1,500,000 times on-load tap-changer operation (based on the number shown in motor-drive unit's counter)	Tap selector maintenance. Please contact Huaming's service department.

Table13.1.3 Maintenance interval



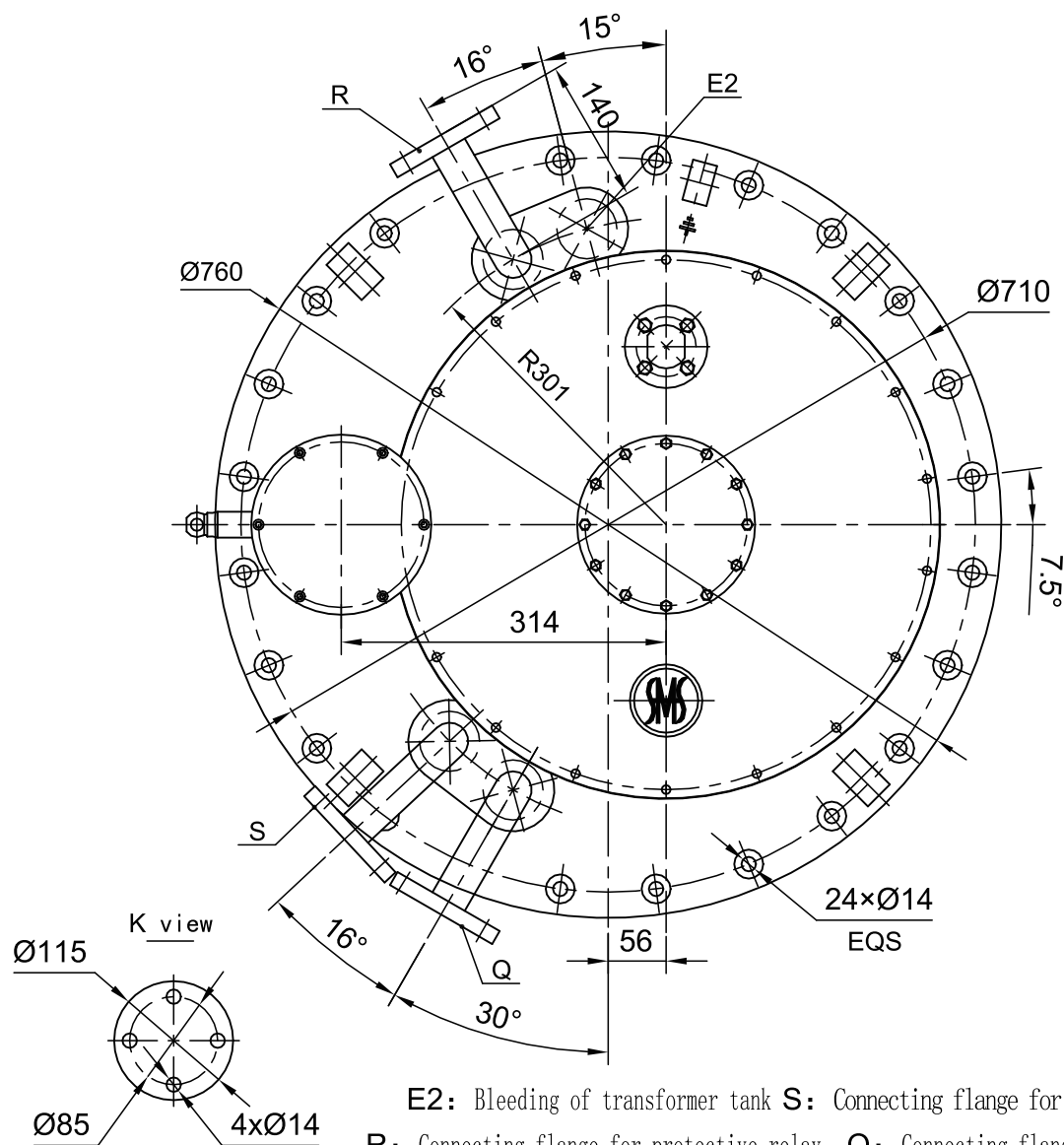
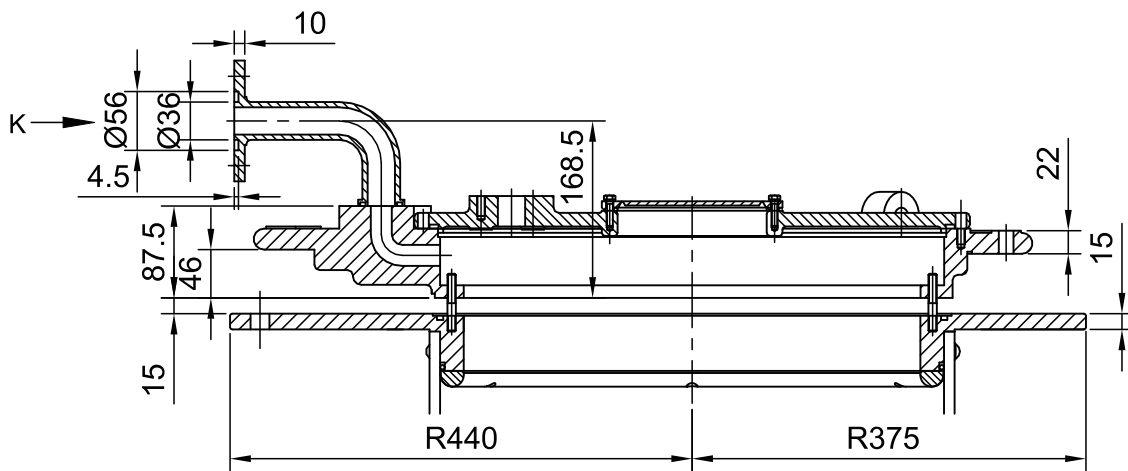
Damage to on-load tap-changer and transformer for non-observance of maintenance interval and improper maintenance.

Please write to export@huaming.com for any support you may need.

14. Annexe

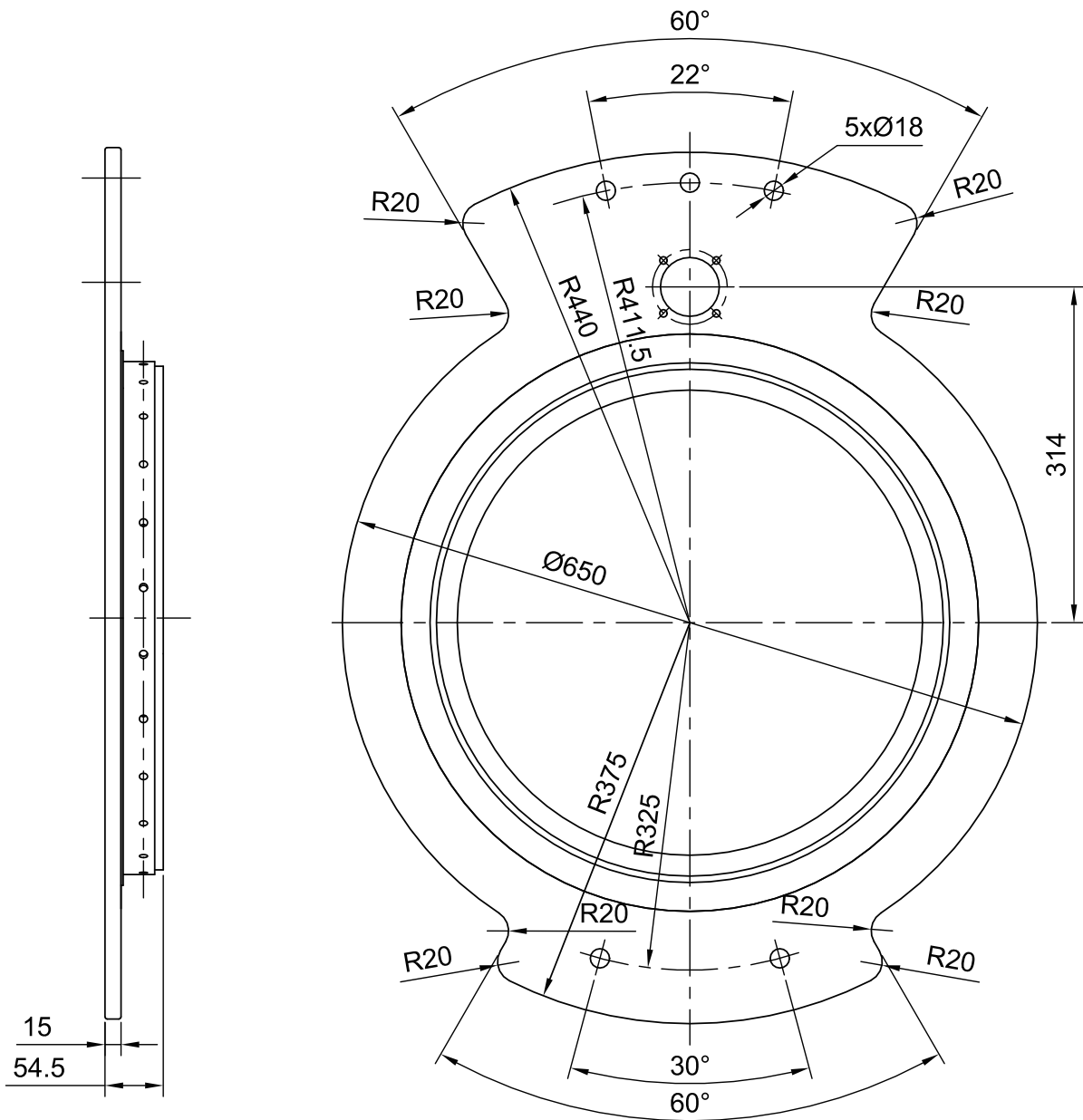
Appendix 1 SHZVG mounting flange dimension	56
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Appendix 1

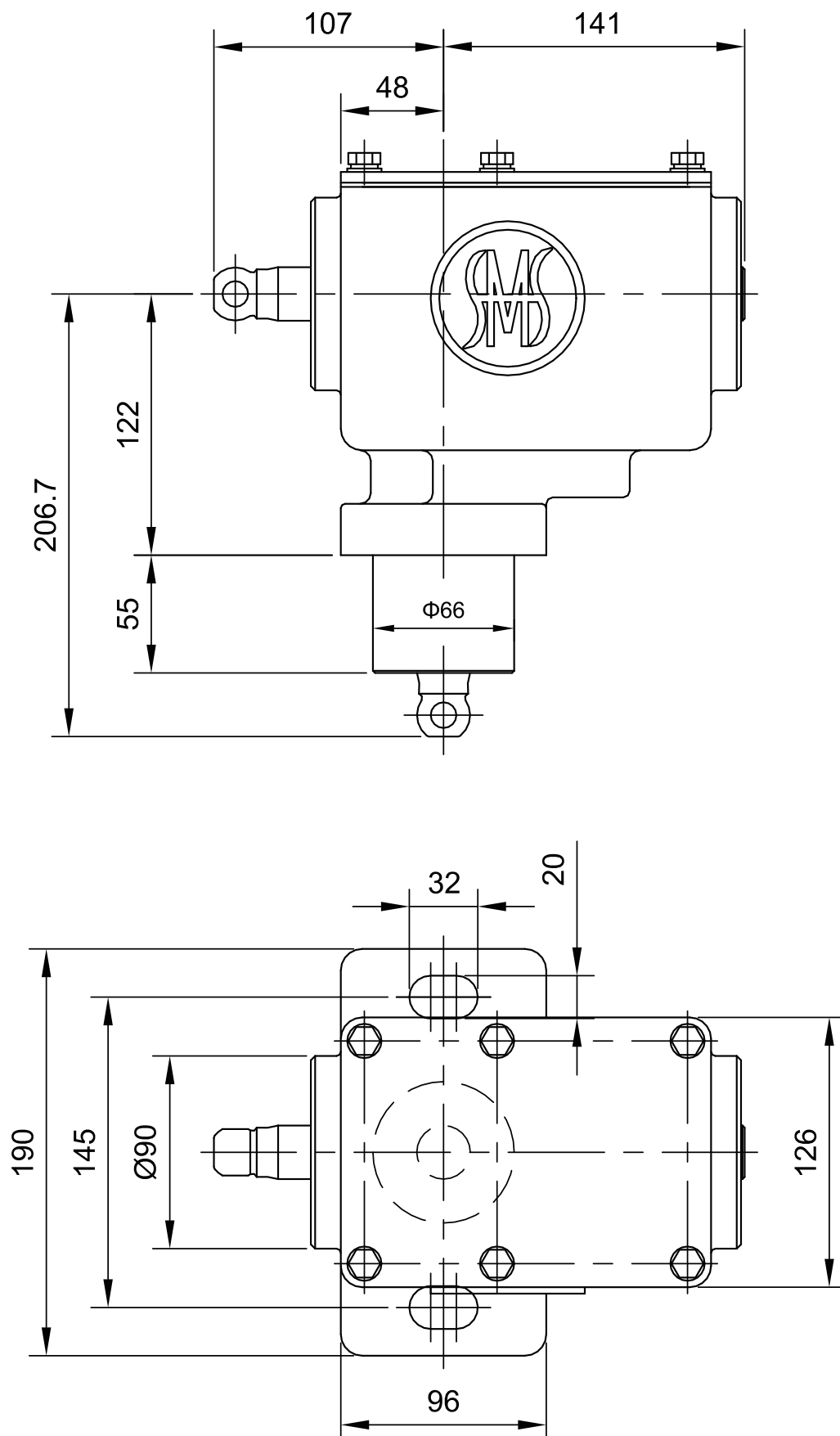


E2: Bleeding of transformer tank S: Connecting flange for suction pipe
R: Connecting flange for protective relay Q: Connecting flange for oil return
Note: Top gear and all pipes can be rotate 360°.

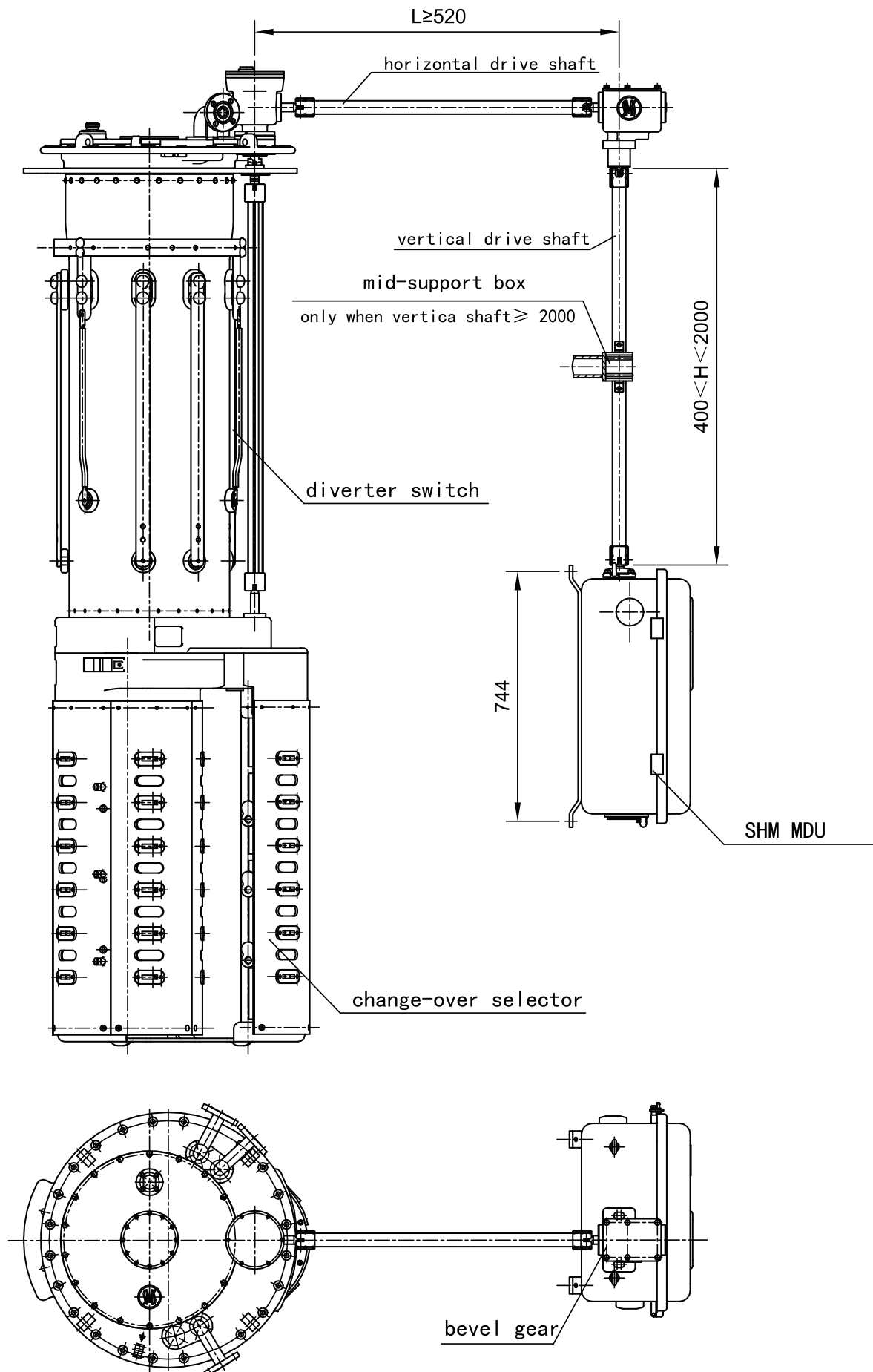
Appendix 3 SHZVG supporting flange dimension



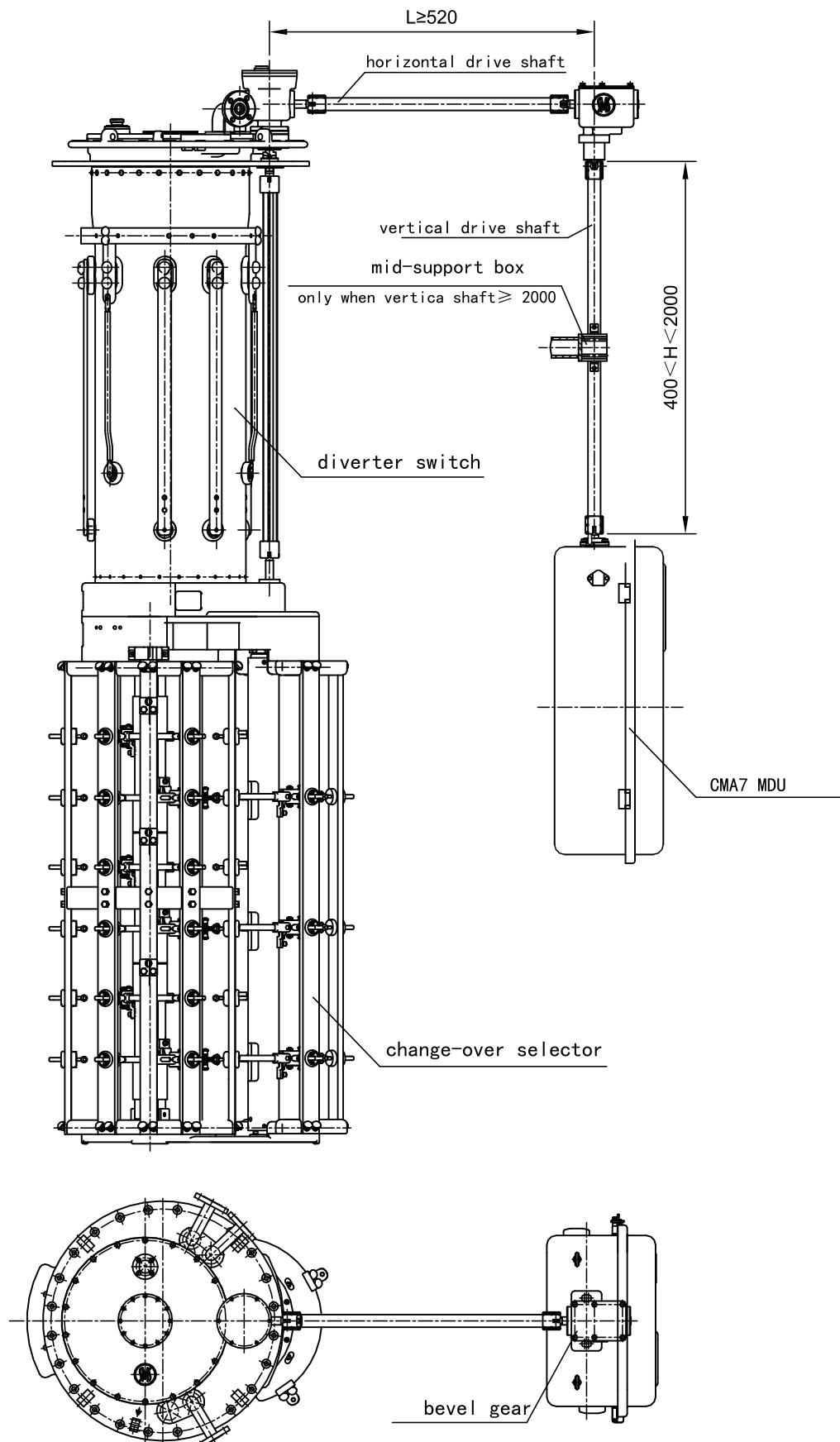
Appendix 4 bevel gear dimension

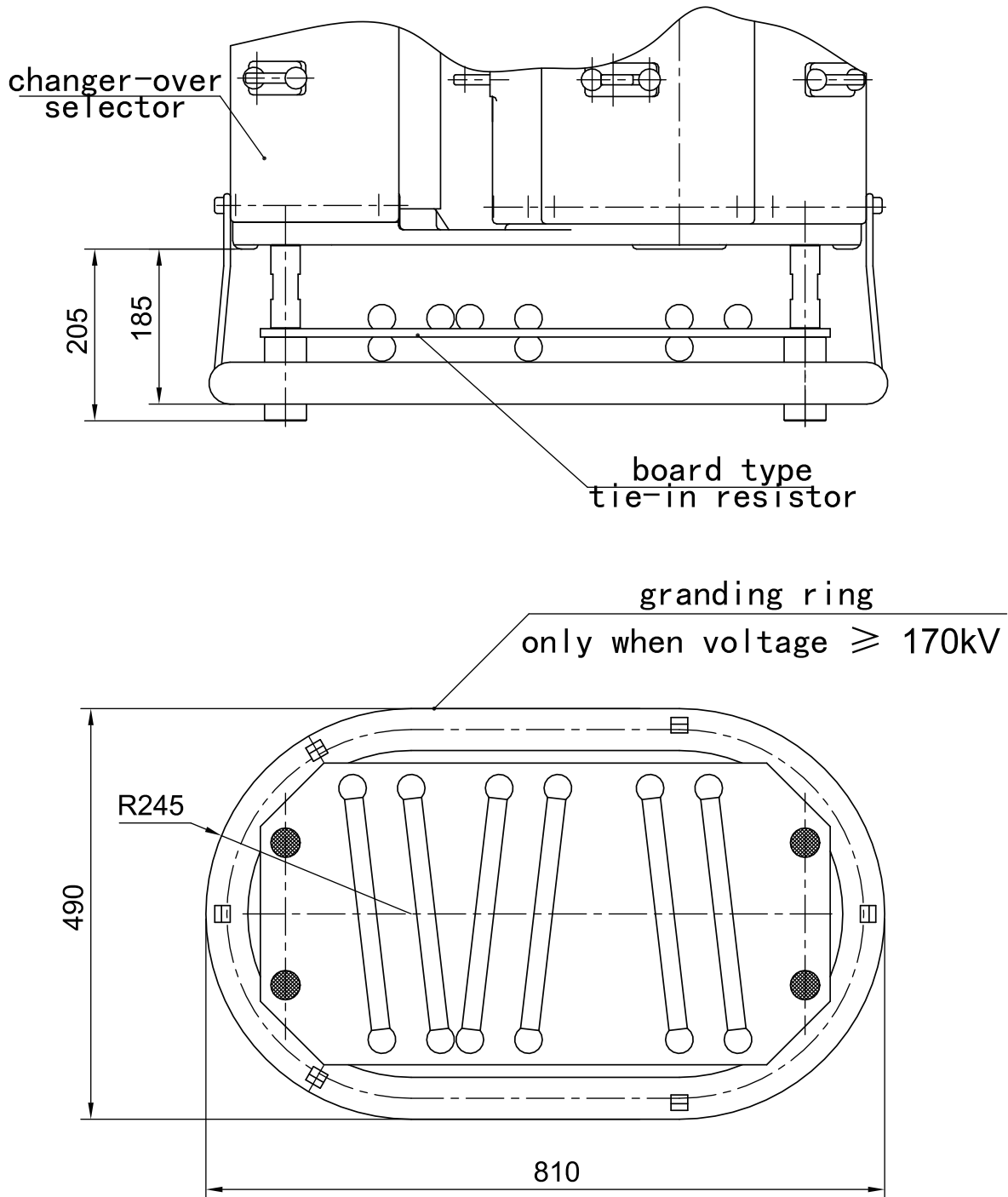


Appendix 5 SHZVG & SHM connection arrangement diagram

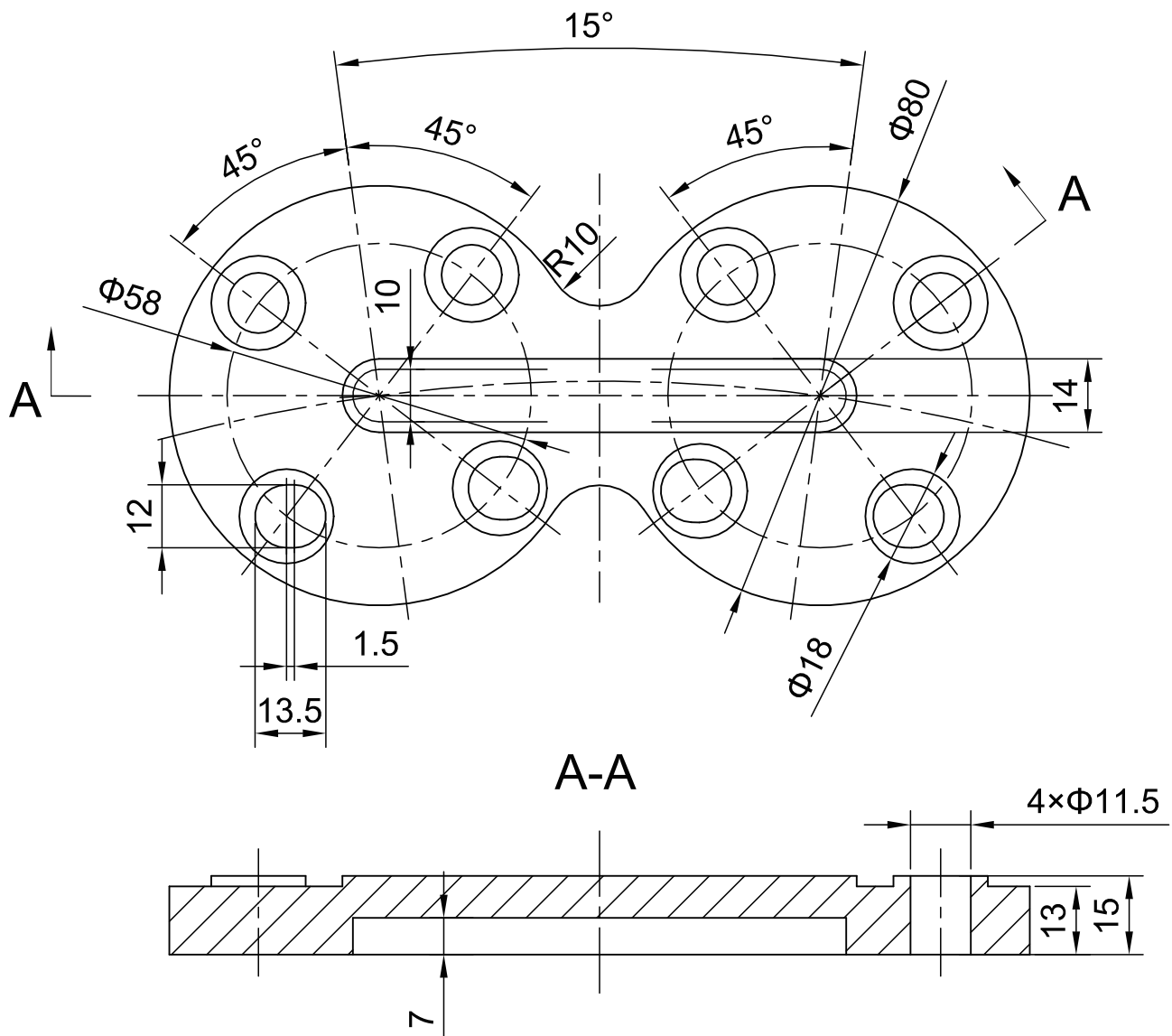


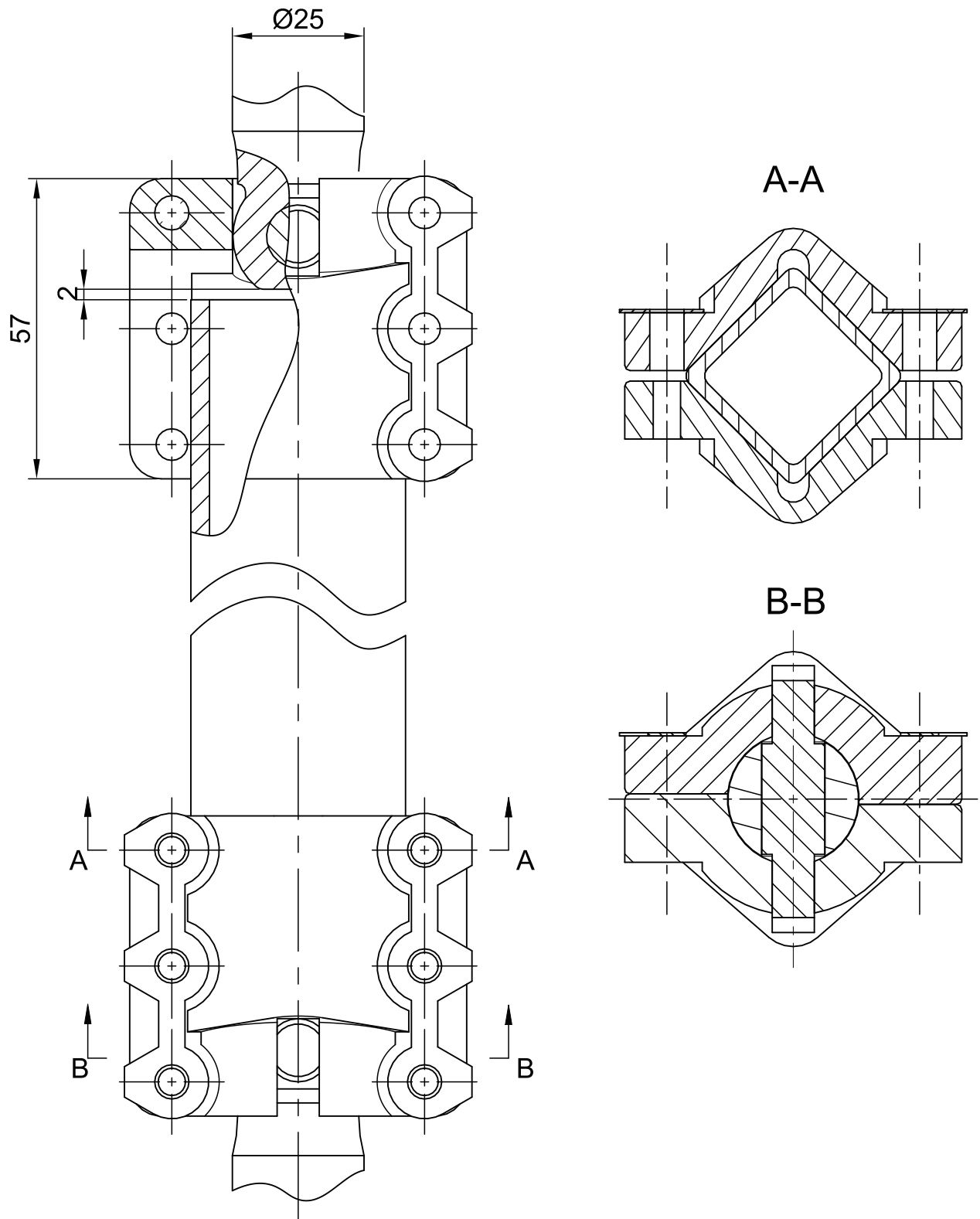
Appendix 6 SHZVG & CMA7 connection arrangement diagram



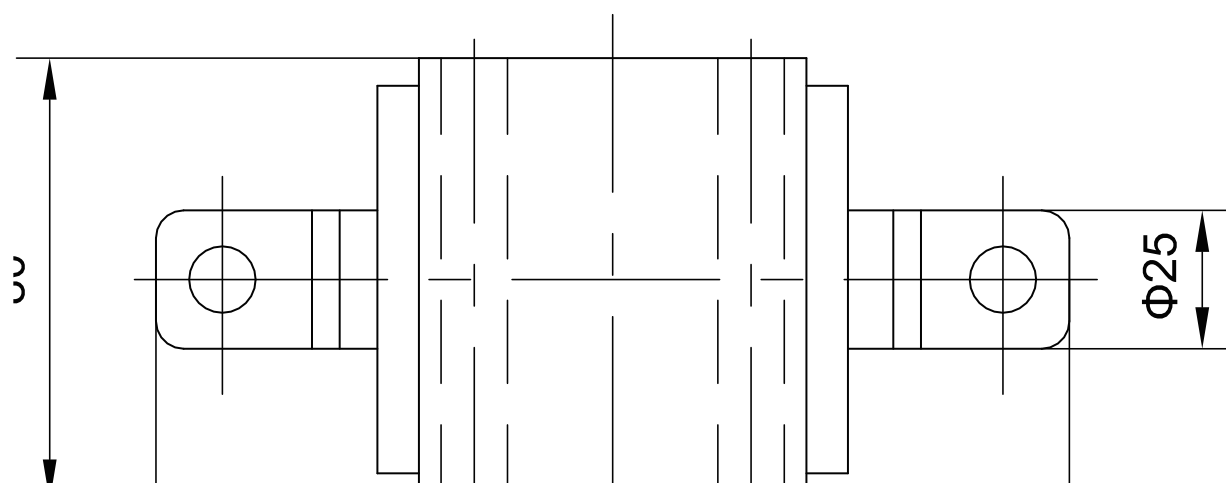
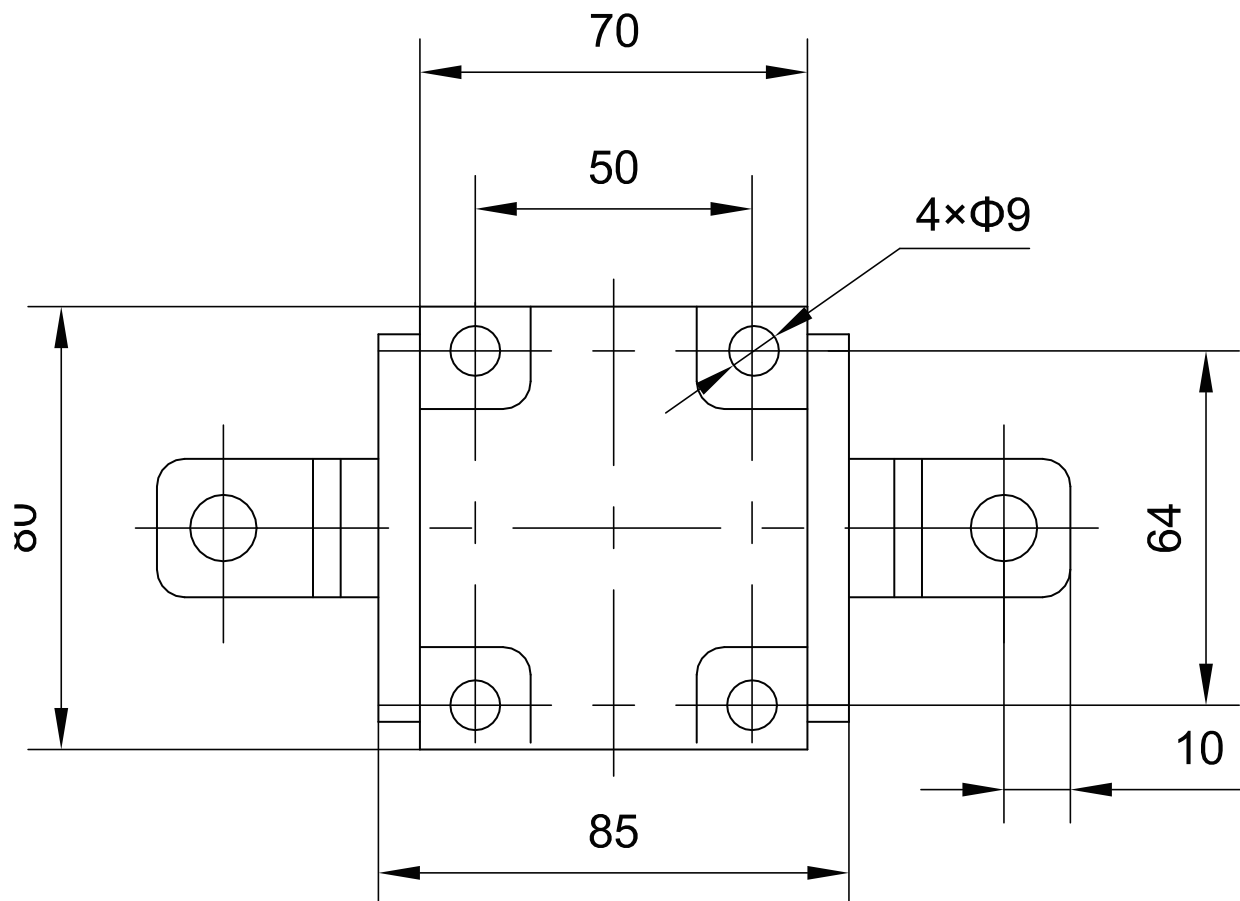
Appendix 7 board type tie-in resistor mounting diagram

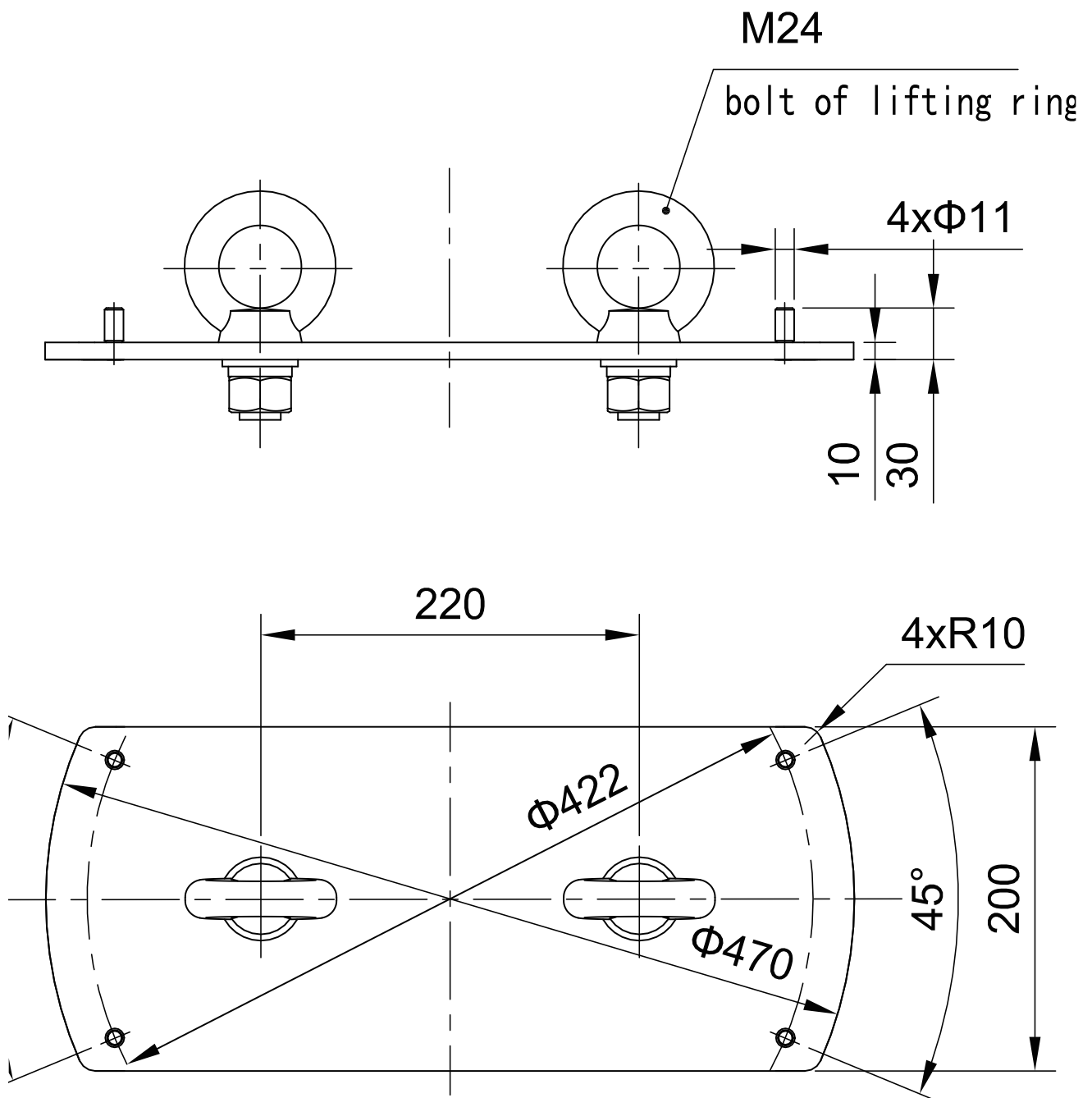
Appendix 8 by-pass pipe



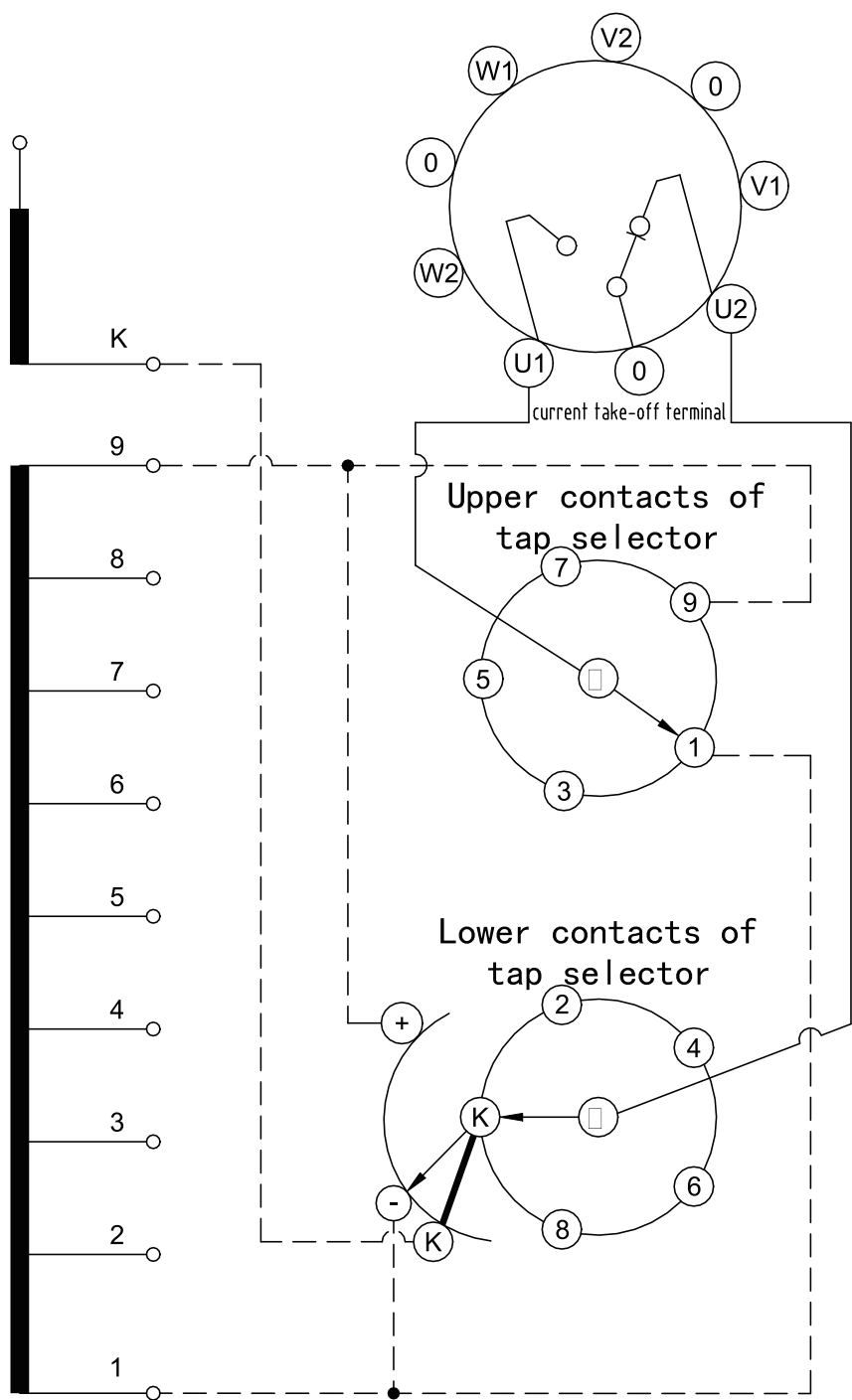
Appendix 9 horizontal & vertical drive shaft

Appendix 10 mid-support bos dimension



Appendix 11 lifting ring mounting dimension

Appendix 12 10193W(T) working position & connection diagram



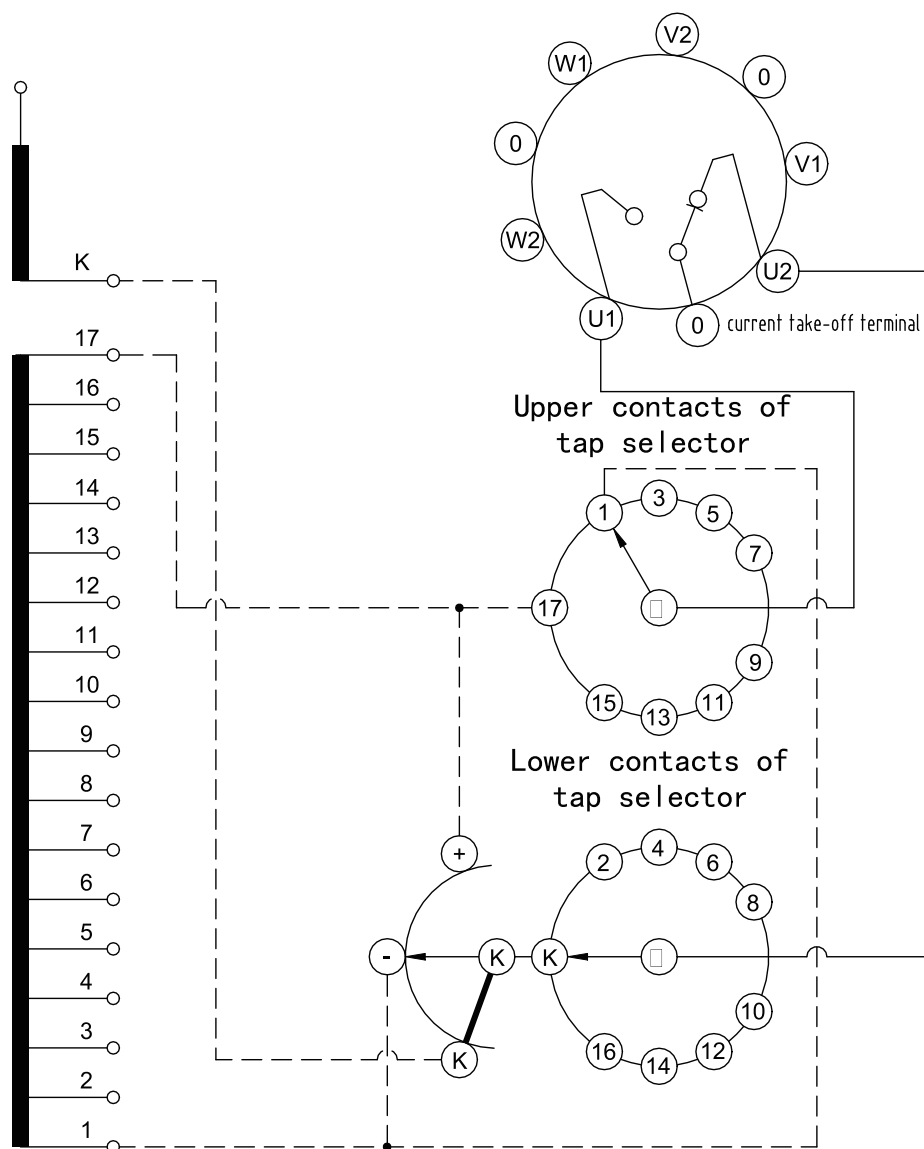
Operation position number	19
Different voltage number	17
Set position	9b●

————— connect by HM
----- connect by user

Change-over selector location	K+										K-									
Tap selector contact position	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9	
Display position	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17	

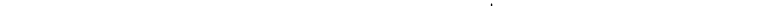
● ← is set position

Appendix 13 18353W(L) working position & connection diagram



Operation position number	35
Different voltage number	33
Set position	17b •

— connect by HM
 --- connect by user

Change-over selector location																																			
Tap selector contact position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17a	17b	17c	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33

• ← — • is set position



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